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MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

054702 - AUTOMATION AND CONTROL LABORATORY

LABORATORY REPORT FOR THE SETUP: SEESAW

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Academic year:
2025-2026

1. Introduction

This project controls a Quanser IP02 Linear Base Unit paired with the Seesaw-E module, a single-input multi-output stabilization problem with both Right-Half Plane (RHP) poles and zeros. The resulting Non-Minimum Phase (NMP) behavior requires careful bandwidth separation and pole placement to balance the seesaw without driving the cart into the track limits.

The report follows the full workflow: Euler–Lagrange modelling, friction/deadzone compensation, hardware identification, classical cascade loop shaping, state-space control with pole placement and LQR/LQI, and a nonlinear lift-up proposal using CasADi trajectory optimization with Time-Varying LQR (TV-LQR).

2. Setups Description

2.1. Quanser IP02 Linear Base Unit

As a baseline, the cart and rail assembly are treated as a standalone, decoupled system isolated from the seesaw dynamics. This isolation permits the independent characterization of linear motion parameters and actuator characteristics. The experimental platform features a DC servo motor that drives a linear cart via a rack-and-pinion mechanism along a horizontal track.

The linear displacement of the cart along the rail is denoted as x_c . The nominal physical constants provided by the manufacturer datasheet are summarized in the table below.

IP02 Linear Base Unit Physical Parameters

Symbol	Description	Value	Unit
R_m	Motor armature resistance	2.6	Ω
k_t	Motor torque constant	7.68×10^{-3}	$N \cdot m/A$
k_m	Back-emf constant	7.68×10^{-3}	$V/(rad/s)$
J_m	Rotor moment of inertia	3.90×10^{-7}	$kg \cdot m^2$
K_g	Planetary gearbox gear ratio	3.71	-
η_m	Motor efficiency	0.69	-
η_g	Gearbox efficiency	0.90	-
M_c	Mass of cart	0.38	kg
M_w	Mass of cart weight	0.37	kg
r_{mp}	Motor pinion radius	6.35×10^{-3}	m
$B_{eq,w}$	Viscous damping (Cart and Weight)	5.4	$N \cdot m \cdot s/rad$

Table 1: Initial datasheet parameters for the Quanser IP02.

2.1.1. Analytical Modeling via Newton's Second Law

To accurately capture the system dynamics, the effective inertial load must account for both the physical mass of the assembly and the reflected rotational inertia of the motor armature. The total equivalent lumped mass M_e translated along the rail is defined as:

$$M_e = M_c + M_w + \frac{J_m K_g^2}{r_{mp}^2} \quad (1)$$

Applying Newton's Second Law to the translational motion of the cart, we account for the driving force F_c and the resistive forces of friction. The equation of motion is:

$$F_c = M_e \cdot \ddot{x}_c + B_{eq} \cdot \dot{x}_c \quad (2)$$

Where:

- M_e is the total equivalent translational mass of the cart assembly.
- B_{eq} is the equivalent linear viscous damping coefficient, capturing the lumped dissipative losses across the track, bearings, and gears.
- F_c is the driving linear force delivered to the rack by the DC motor pinion.

2.1.2. Actuator Dynamics and Electromechanical Coupling

The driving force F_c originates from the electrical domain, where the input armature voltage V_m generates an electromagnetic torque. Accounting for back-EMF and armature resistance, the force is defined as:

$$\begin{aligned} F_c &= \frac{\eta_g K_g k_t}{R_m r_{mp}} \left(\eta_m \cdot v_m - \frac{K_g k_m}{r_{mp}} \cdot \dot{x}_c \right) \\ &= \alpha_f \eta_m \cdot v_m - B_{emf} \cdot \dot{x}_c \end{aligned} \quad (3)$$

Expanding this expression allows the electrical and mechanical dependencies to be cleanly decoupled into a voltage-proportional force gain and an intrinsic electrical damping term, namely $\alpha_f \eta_m$ and B_{emf} .

2.1.3. Equation of Motion

Substituting the actuator force expression into the mechanical force balance yields the explicit differential equation mapping the control input voltage $V_m(t)$ directly to the cart linear acceleration $\ddot{x}_c(t)$:

$$\alpha_f \eta_m \cdot v_m(t) - B_{emf} \cdot \dot{x}_c(t) = M_e \cdot \ddot{x}_c(t) + B_{eq,c} \cdot \dot{x}_c$$

Grouping the velocity-dependent terms highlights the total open-loop damping present within the system:

$$\alpha_f \eta_m \cdot v_m(t) = M_e \cdot \ddot{x}_c(t) + B_{tot} \cdot \dot{x}_c(t) \quad (4)$$

Where $B_{tot} = B_{eq} + B_{emf}$ represents the combined effective damping coefficient, encapsulating both mechanical friction losses and back-EMF braking effects. This consolidated linear model serves as the analytical foundation for subsequent system identification and control loop synthesis.

2.2. Quanser Seesaw-E module plus IP02

In addition to the previous setup, the IP02 cart is mounted onto the Seesaw-E frame. The system's complexity increases due to the cross-coupling between the translational and rotational subsystems. The cart's linear displacement (x_c) now acts as a time-varying shifting mass that generates a dynamic torque about the pivot, influencing the seesaw's angular position (θ). Conversely, the angular tilting of the seesaw introduces gravitational components along the track axis, modulating the translational forces acting on the cart.

The nominal technical specifications and physical parameters for the integrated Seesaw-E module are outlined below:

SEESAW-E System Specifications

Symbol	Description	Value	Unit
M_{SW}	Mass (Seesaw + IP02 Track)	3.6	kg
K_{gs}	Geartrain Gear Ratio	3	-
D_T	Pivot to IP02 Track Distance	0.125	m
$D_{\hat{C}}$	Pivot to Center of Gravity Distance	0.058	m
J_{SW}	Moment of Inertia (about CoG)	0.395	$kg \cdot m^2$
B_{SW}	Viscous Damping Coefficient	≈ 0	N-m-s/rad
g	Gravitational Constant	9.81	m/s^2
K_{E_SW}	Encoder Resolution	0.0015	rad/count
e_{range}	Angular Range (on flat surface)	± 11.5	$^\circ$

Table 2: Technical specifications for the Quanser SEESAW-E module.

2.2.1. Analytical Modeling via Euler-Lagrange Equation

To model the multi-degree-of-freedom (MDOF) coupled system, the analytical derivation uses the energy-based Euler-Lagrange method. We select the generalized coordinates vector as:

$$q = \begin{bmatrix} x_c \\ \theta \end{bmatrix} \quad (5)$$

First, considering the baseline moment of inertia of the seesaw assembly about its physical pivot point, we define J_{pivot} as:

$$J_{pivot} = J_{SW} + M_C \cdot D_C^2$$

The total kinetic energy T of the coupled system accounts for the linear translation of the cart, the pure rotation of the seesaw base, and the non-linear velocity coupling terms arising from the moving reference frame:

$$T = \frac{1}{2}M_e \cdot \dot{x}_c^2 + \frac{1}{2}(J_{pivot} + M_{tot}D_T^2) \cdot \dot{\theta}^2 + \frac{1}{2}M_{tot} \cdot x_c^2 \cdot \dot{\theta}^2 - M_{tot}D_T \cdot \dot{x}_c \cdot \dot{\theta} \quad (6)$$

Where $M_{tot} = M_C + M_w$ represents the total combined active mass of the translating assembly.

The system potential energy V is defined relative to the horizontal datum plane, capturing the height variations of both the translating cart mass along the inclined rail and the shift in the center of gravity of the rotating seesaw frame:

$$V = gM_{tot} \cdot x_c \cdot \sin(\theta) + g(M_{tot}D_T + M_{SW}D_C) \cdot \cos(\theta) \quad (7)$$

Using these energy definitions, the system Lagrangian is established as $L = T - V$. The explicit equations of motion are generated by applying the Euler-Lagrange operator to each generalized coordinate:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = Q_i \quad (8)$$

Where Q_i represents the vector of generalized non-conservative forces and dissipative torques acting on the system. For the translational and rotational coordinates respectively, the formulation expands to:

$$\begin{cases} \frac{\partial^2 L}{\partial t \partial \dot{x}_c} - \frac{\partial L}{\partial x_c} = -B_{tot} \cdot \dot{x}_c + \alpha_f \eta_m \cdot v_m \\ \frac{\partial^2 L}{\partial t \partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = -B_{SW} \cdot \dot{\theta} \end{cases} \quad (9)$$

By evaluating the partial differentiations explicitly, this system yields a set of coupled, non-linear second-order differential equations that describe the cart-seesaw dynamics. These equations serve as the analytical model for subsequent linearization, identification, and model-based controller design.

2.2.2. Equation of Motion

By explicitly evaluating the Euler-Lagrange equations, we obtain the coupled, non-linear single-input multi-output (SIMO) dynamic equations for the complete cart-seesaw system. The resulting system models the full electromechanical translation of the cart alongside the rotational dynamics of the supporting seesaw frame:

$$\begin{cases} M_e \cdot \ddot{x}_c - B_{tot} \cdot \dot{x}_c - M_{tot}D_T \cdot \ddot{\theta} - M_{tot} \cdot x_c \cdot \dot{\theta}^2 + gM_{tot} \cdot \sin(\theta) = \alpha_f \eta_m \cdot v_m \\ M_{tot} \cdot \ddot{\theta} \cdot x_c^2 + 2M_{tot} \cdot \dot{x}_c \cdot \dot{\theta} + gM_{tot} \cdot \cos(\theta) \cdot x_c - M_{tot}D_T \cdot \ddot{x}_c + \\ (J_{SW} + M_{tot}D_T^2) \cdot \ddot{\theta} + B_{sw} \cdot \dot{\theta} - g(M_{tot}D_T + M_{SW}D_C) \cdot \sin(\theta) = 0 \end{cases} \quad (10)$$

Where:

- $V_m(t)$ is the control input voltage applied to the linear base unit motor, standardizing the actuator parameters derived previously.
- The centripetal acceleration component $M_{tot} \cdot x_c \cdot \dot{\theta}^2$ and the Coriolis coupling term $2M_{tot} \cdot \dot{x}_c \cdot \dot{\theta}$ represent the highly non-linear dynamic forces acting across the moving reference frames.

2.2.3. Linearization

Because these coupled non-linear equations are unsuitable for linear controller design, the system must be linearized around its unstable equilibrium point. We define the nominal operating equilibrium where the cart rests at the center of the track and the seesaw is horizontal:

$$(x_{c,0}, \theta_0) = (0, 0)$$

To perform a small-signal linearization around this equilibrium state, the following assumptions are enforced:

1. Small angles ($\theta \simeq 0$) : $\sin(\theta) = \theta$ and $\cos(\theta) = 1$

2. Small displacement ($x_c \simeq 0$)

3. Ignoring high-order velocity terms : $\dot{\theta}^2, x_c^2, \dot{x}_c \dot{\theta} \simeq 0$

Applying these assumptions removes the non-linear coupling terms and simplifies the model into a set of linearly coupled differential equations. This linearized model describes the small-signal perturbations of the system near the origin:

$$\begin{cases} M_e \cdot \ddot{x}_c + B_{tot} \cdot \dot{x}_c \\ -M_{tot} D_T \cdot \ddot{\theta} + g M_{tot} \cdot \theta = \alpha_f \eta_m \cdot v_m \\ -M_{tot} D_T \cdot \ddot{x}_c + g M_{tot} \cdot x_c \\ +(J_{sw} + M_{tot} D_T^2) \cdot \ddot{\theta} + B_{sw} \cdot \dot{\theta} - g(M_{tot} D_T + M_{sw} D_C) \cdot \theta = 0 \end{cases} \quad (11)$$

This linear differential model forms the analytical model used for transfer-function derivation, state-space representation, and closed-loop controller design throughout the remainder of this report.

2.3. Non-linearities Mitigation

During the experimental validation of the IP02 linear base unit, discrepancies emerged between the idealized linear models and the physical hardware response. These discrepancies are primarily driven by localized, hard non-linearities, specifically static friction, Coulomb friction, and actuator deadzones inherent to the rack-and-pinion and planetary gearbox assembly.

Left uncompensated, these non-linearities introduce steady-state tracking errors, limit cycles, and severe phase distortions that degrade linear controller performance. To restore the validity of the linear assumptions, a data-driven identification routine was used to characterize these nonlinear effects, followed by the implementation of a software-based feedforward inverse-deadband compensation scheme.

The underlying mathematical model for this non-symmetric voltage deadzone is formulated as:

$$V_{eff} = f(V_{cmd}) = \begin{cases} V_{cmd} + u_{d,pos} & \text{if } V_{cmd} > \epsilon \\ 0 & \text{if } |V_{cmd}| \leq \epsilon \\ V_{cmd} - u_{d,neg} & \text{if } V_{cmd} < -\epsilon \end{cases} \quad (12)$$

Where:

- ϵ is the deadzone threshold, used to avoid chattering between positive and negative motion.

- V_{cmd} is the raw voltage command generated by the controller.
- V_{eff} is the effective, torque-producing voltage delivered to the motor armature after overcoming friction.
- $u_{d,pos}$ and $u_{d,neg}$ represent the distinct positive and negative voltage thresholds required to break static friction.

2.3.1. Characterization via Dynamic Modular Signaling

To cleanly decouple velocity-dependent Coulomb friction from zero-velocity stiction, the system was excited using a structured modular input signal designed to sweep through multiple distinct dynamic phases. This drives the cart through static equilibrium, acceleration, and steady-state velocity regimes, which exposes the non-symmetric voltage deadzone.

2.3.2. Algorithmic Deadzone Extraction

The script `seesaw_nonlinear_model.m` filters the encoder position, estimates velocity, masks sustained-motion intervals, and extracts separate positive and negative breakaway voltages from the minimum command that produces motion in each direction. The resulting thresholds are used only as feedforward compensation; the residual ripple in Fig. 1 shows that the track friction is not fully captured by a static deadzone model.

The extraction is intentionally simple: zero-phase filtering avoids adding phase lag before differentiation; a velocity threshold of $|\dot{x}_c| > 0.002$ m/s rejects encoder ticks and quasi-static samples; the remaining positive and negative motion windows are split to estimate $u_{d,pos}$ and $u_{d,neg}$. This keeps the identified compensation tied to measured motion rather than to isolated voltage spikes.

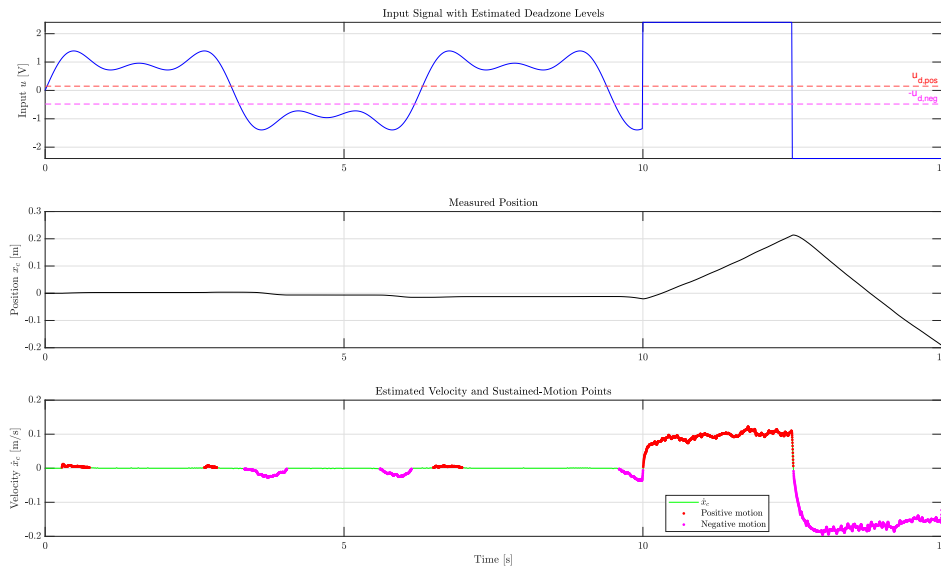


Figure 1: Deadzone Estimation results generated by the extraction script. Top: Input voltage signal highlighting the identified deadzone limits ($u_{d,pos}$ and $u_{d,neg}$). Middle: Filtered cart position. Bottom: Estimated cart velocity with overlaid markers indicating points of validated sustained motion.

2.3.3. Software-Based Inverse-Deadband Compensation

Using the extracted parameters stored in `param_nonlinear.mat`, an online feed-forward compensation block was placed directly upstream of the Digital-to-Analog Converters (DAC). This software block applies an algebraic inverse mapping function to the controller voltage output (V_{ctrl}) before routing the signal to the power amplifier:

$$V_{DAC} = \begin{cases} V_{ctrl} + u_{d,pos} & \text{if } V_{ctrl} > \epsilon \\ 0 & \text{if } V_{ctrl} = 0 \\ V_{ctrl} - u_{d,neg} & \text{if } V_{ctrl} < -\epsilon \end{cases} \quad (13)$$

The compensation pre-shifts the command near the zero crossing, reducing the voltage lost to stiction. It improves the usability of the linear model but does not remove the remaining friction mismatch, which remains visible in the later time and frequency validation.

3. Dynamic Modeling & System Identification

3.1. Cart Analytical Modeling: Transfer Function Definition

Based on the equations of motion derived in sec. 2.1.3, we define the open-loop transfer function of the isolated cart system. Applying the Laplace transform under the assumption of zero initial conditions yields the transfer function mapping the motor armature voltage $V_m(s)$ to the cart position $X_c(s)$:

$$\begin{aligned} \alpha_f \eta_m \cdot V_m(s) &= [M_e \cdot s^2 + B_{tot} \cdot s] \cdot X_c(s) \\ G_x(s) &= \frac{X_c(s)}{V_m(s)} = \frac{\alpha_f \eta_m}{M_e \cdot s^2 + B_{tot} \cdot s} \end{aligned} \quad (14)$$

Because the position transfer function contains a pure integrator, it exhibits an infinite static gain. Consequently, the identification procedure uses the velocity transfer function, $G_v(s)$, which forms a stable, first-order LTI system:

$$G_v(s) = \frac{s \cdot X_c(s)}{V_m(s)} = \frac{\alpha_f \eta_m}{M_e \cdot s + B_{tot}} \quad (15)$$

This velocity plant features a single stable pole located at $p_{cart} = -B_{tot}/M_e$. Its fundamental dynamic characteristics are defined by its steady-state DC gain, K_{DC} , and its time constant, τ :

$$K_{DC} = \frac{\alpha_f \eta_m}{B_{tot}}, \quad \tau = \frac{M_e}{B_{tot}} \quad (16)$$

3.2. Cart Parameter Estimation & Open-Loop Time Validation

While nominal parameters provide a strong theoretical baseline, physical hardware introduces discrepancies due to mechanical wear, dust accumulation in the bearings, and unmodeled frictional variations. A data-driven parameter estimation routine was therefore executed to improve the cart model used for controller design.

The focus of this tuning is placed strictly on the equivalent viscous damping coefficient (B_{eq}) and the planetary gearbox efficiency (η_g), as mechanical param-

eters are highly susceptible to degradation compared to stable electromagnetic constants (like R_m and η_m).

3.2.1. Methodology

The system was excited on the hardware using a 2.0 V step input. To accurately isolate the velocity states without introducing numerical phase lag, the raw encoder position data was differentiated in post-processing. A zero-phase, 2nd-order Butterworth filter (`filtfilt` in MATLAB) was applied, dynamically tuned relative to the theoretical structural bandwidth ($\omega_c = 2 \cdot B_{tot}/M_e$).

Parameter identification was formulated as an unconstrained non-linear optimization problem. Using the native MATLAB minimization algorithm (`fminsearch`), the routine minimizes a cost function defined by the Mean Squared Error (MSE) between the measured hardware velocity and the simulated response:

$$\min_{\eta_g, B_{eq}} J = \frac{1}{N} \sum_{k=1}^N (\dot{x}_{c,hw}(k) - \dot{x}_{c,sim}(k))^2 + \Phi(\eta_g) \quad (17)$$

Where $\Phi(\eta_g)$ corresponds to a barrier penalty acting as a soft constraint, bounding the gearbox efficiency between 50% and 110% of its nominal datasheet value to keep the estimate physically realistic. This local optimization reduces the measured velocity error and fixes the refined values of B_{eq} and η_g for subsequent controller syntheses.

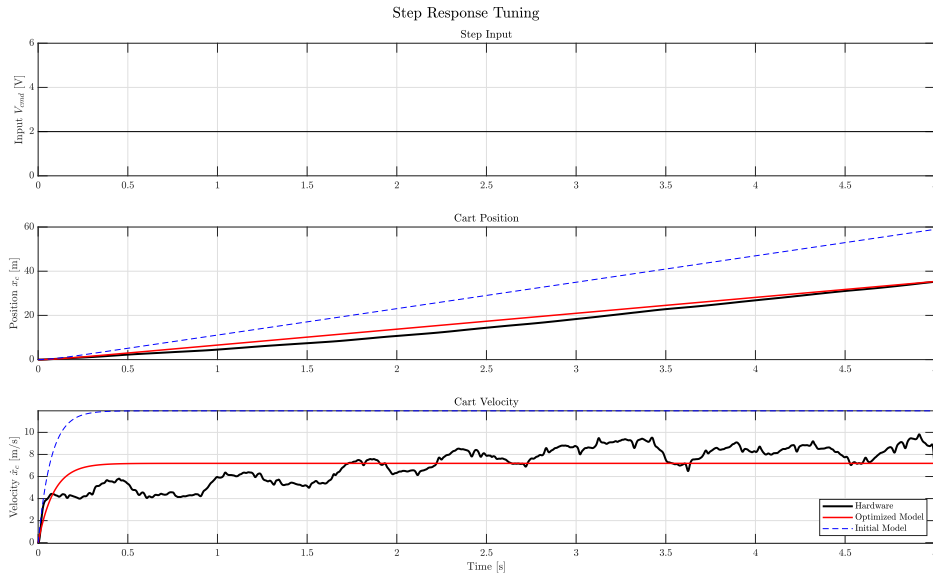


Figure 2: Step Response Tuning. The plot compares the raw hardware telemetry (black) against the initial nominal model (blue dashed) and the final `fminsearch` optimized model (red).

The final identified values are $B_{eq,w} = 6.7217$ Ns/m and $\eta_g = 0.729$. These values replace the nominal datasheet values $B_{eq,w} = 5.4$ Ns/m and $\eta_g = 0.90$ in the subsequent cart transfer-function definition and controller design. With these tuned parameters, the resulting total damping becomes $B_{tot} = 12.3669$ Ns/m.

As shown in Fig. 2, the optimized model follows the measured hardware response

more closely than the initial nominal model, especially in the transient velocity response. Nevertheless, the match is not perfect: the measured velocity still contains small ripples and local deviations that are not reproduced by the linear first-order model. This behaviour is consistent with track irregularities, Coulomb friction, stiction, and velocity-dependent friction effects in the rack-and-pinion mechanism. Therefore, the tuned model is accurate enough for controller design, while the remaining mismatch is treated as an expected hardware nonideality to be handled by conservative validation and feedback control.

3.3. Cart Open-Loop Model Frequency Validation

To design a stable controller for the seesaw cascade, the inner-loop cart model must be accurately characterized across a wide frequency spectrum.

3.3.1. Methodology: Schroeder-Phased Multi-Sine Injection

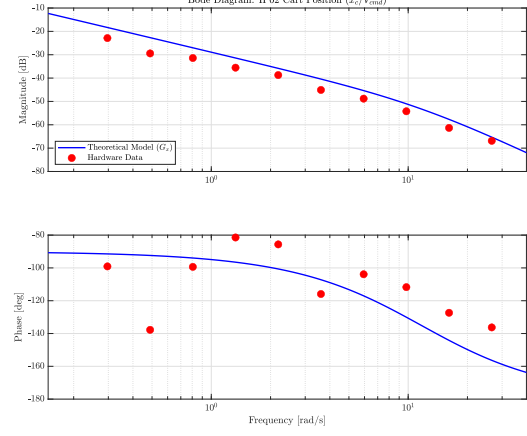
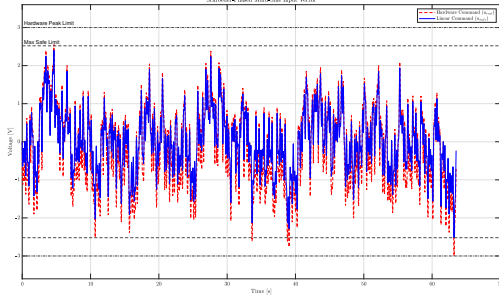
A Schroeder-phased multi-sine was used instead of a chirp so that $N = 10$ logarithmically spaced frequencies could be tested in one safe hardware run. The phases $\phi_k = -\pi k(k-1)/N$ reduce the crest factor of the combined signal, allowing more usable energy at each target frequency without exceeding the voltage budget. The generated command was then checked against peak voltage, current, track travel, and rack-force limits before execution.

3.3.2. Dynamic Frequency Limits and Safety Checklist

The lower frequency bound avoids end-stop collisions by limiting the predicted displacement to 75% of the half-track length. The upper bound begins from the required inner-loop bandwidth target but is capped when the predicted rack force approaches the empirical clicking limit. The final pre-flight simulation checked displacement, rack force, peak current below 3.0 A, and RMS current below 1.0 A before the signal was deployed.

3.3.3. Digital Signal Processing and Spectral Extraction

Following execution on the hardware, the transient part was trimmed and the steady segment was detrended and windowed before estimating the empirical magnitude and phase at the injected frequencies. Evaluating only the injected tones makes the extraction behave like a narrow-band lock-in measurement: broadband encoder noise and unmodelled disturbances are largely rejected, while the Bode points remain directly comparable to the tuned $G_x(s)$.



(a) Safe Schroeder multi-sine command with inverse-deadband shift.

(b) Extracted FRF against the tuned cart model.

Figure 3: Open-loop cart frequency-identification signal and extracted empirical Bode validation.

The identified model is considered an adequate fit: although it slightly overestimates the magnitude, it follows the correct trend in both magnitude and phase across the swept band. The residual mismatch, including the larger phase deviation near 0.5 rad/s, is attributed to the same track inconsistencies visible in the step-response tuning (fig. 2), which introduce friction-dependent behaviour that the linear model does not reproduce.

3.4. Coupled Cart-Seesaw Analytical Modeling: Transfer Function Definition

To analyze the internal dynamics of the outer loop (angle control), we derive the transfer function mapping the cart's position x_c to the seesaw's angle θ , expressed at eq. 10. Deriving the Laplace transform from the linearized coupled equations of motion (eq. 11), while keeping the zero initial conditions assumption, yields the following matrix relationship:

$$\begin{bmatrix} M_e \cdot s^2 + B_{tot} \cdot s & -M_{tot}D_T \cdot s^2 + gM_{tot} \\ -M_{tot}D_T \cdot s^2 + gM_{tot} & (J_{pivot} + M_{tot}D_T^2) \cdot s^2 + B_{SW} \cdot s - g(M_{tot}D_T + M_{SW}D_C) \end{bmatrix} \begin{bmatrix} X_c(s) \\ \Theta(s) \end{bmatrix} = \begin{bmatrix} \alpha_f \eta_m \\ 0 \end{bmatrix} V_m(s) \quad (18)$$

By isolating the second equation, which represents the purely rotational dynamics of the seesaw base, we extract the explicit transfer function $G_\theta(s)$ mapping the cart displacement to the angular tilt:

$$G_\theta(s) = \frac{\Theta(s)}{X_c(s)} = \frac{M_{tot}D_T \cdot s^2 - gM_{tot}}{(J_{pivot} + M_{tot}D_T^2) \cdot s^2 + B_{SW} \cdot s - g(M_{tot}D_T + M_{SW}D_C)} \quad (19)$$

This transfer function structurally highlights the system's inherent Non-Minimum Phase (NMP) behavior, dictated by the simultaneous presence of a Right-Half Plane (RHP) pole and a RHP zero resulting from the $-gM_{tot}$ term in the numerator.

3.5. State-Space Derivation

For modern control techniques, the eq. 11's system is represented in continuous-time state-space format. We define the standard 4-state vector as $x = [x_c, \theta, \dot{x}_c, \dot{\theta}]^T$.

3.5.1. Matrix Rewriting and Decoupling

Rather than computing the highly coupled second-order derivatives manually, the system is decoupled by defining an effective inertia matrix, M_{eff} , alongside the state constraint matrix A_{rhs} and input vector B_{rhs} :

$$\begin{aligned} M_{eff} &= \begin{bmatrix} M_e & -M_{tot}D_T \\ -M_{tot}D_T & J_{pivot} + M_{tot}D_T^2 \end{bmatrix} \\ A_{rhs} &= \begin{bmatrix} 0 & -gM_{tot} & -B_{tot} & 0 \\ -gM_{tot} & g(M_{tot}D_T + M_{SW}D_C) & 0 & -B_{SW} \end{bmatrix} \\ B_{rhs} &= \begin{bmatrix} \alpha_f \eta_m \\ 0 \end{bmatrix} \end{aligned} \quad (20)$$

The second-order accelerations are computationally resolved via matrix inversion in MATLAB:

$$\begin{bmatrix} \ddot{x}_c \\ \ddot{\theta} \end{bmatrix} = M_{eff}^{-1} A_{rhs} \begin{bmatrix} x_c \\ \theta \\ \dot{x}_c \\ \dot{\theta} \end{bmatrix} + M_{eff}^{-1} B_{rhs} V_m \quad (21)$$

This mathematical formulation allows the continuous-time state-space matrices ($A_{SW}, B_{SW}, C_{SW}, D_{SW}$) to be extended programmatically, avoiding arithmetic truncation errors:

$$A_{SW} = \begin{bmatrix} 0_{2 \times 2} & I_{2 \times 2} \\ M_{eff}^{-1} A_{rhs} & \end{bmatrix}, \quad B_{SW} = \begin{bmatrix} 0_{2 \times 1} \\ M_{eff}^{-1} B_{rhs} \end{bmatrix} \quad (22)$$

Because only the position x_c and the angle θ are directly measurable via the encoders, the output and feedthrough matrices are strictly defined as:

$$C_{SW} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad D_{SW} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (23)$$

3.5.2. Augmented System for Integral Action

To ensure zero steady-state tracking error and robust disturbance rejection during hardware implementation, the standard 4-state model is augmented with a virtual fifth state representing the integral of the seesaw angle:

$$\xi = \int \theta dt \quad (24)$$

The augmented state vector becomes $\eta = [x_c, \theta, \dot{x}_c, \dot{\theta}, \xi]^T$, yielding the new matrices A_5 and B_5 used later:

$$A_5 = \begin{bmatrix} A_{SW} & 0_{4 \times 1} \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}, \quad B_5 = \begin{bmatrix} B_{SW} \\ 0 \end{bmatrix} \quad (25)$$

3.6. Open-Loop Analysis

The linearized cart-seesaw model is open-loop unstable but suitable for state-feedback design. The tuned four-state plant has one right-half-plane pole and one non-minimum-phase zero, so the controller must stabilize the falling mode

without asking the cart to move faster than the inverse response permits. The checks used before controller synthesis are summarized in Table 3.

Property	Result	Design consequence
Stability	One RHP pole; representative deployed PP model poles $\{+2.74, -1.48 \pm j1.30, -14.13\}$ rad/s.	Any balancing controller must move the unstable pole into the LHP before hardware testing.
Controllability	$\text{rank } \mathcal{C}(A_{SW}, B_{SW}) = 4/4$ and $\text{rank } \mathcal{C}(A_5, B_5) = 5/5$.	Both full-state feedback and integral augmentation are controllable.
Observability	With measured x_c, θ , $\text{rank } \mathcal{O}(A_{SW}, C_{SW}) = 4/4$; the augmented LQI design checks 5/5.	Velocity states can be reconstructed in principle, but estimator bandwidth is limited by encoder quantization.
Operating range	Track travel is limited to about ± 0.432 m and the seesaw mechanical range is about $\pm 11.5^\circ$.	References and validation signals are tapered or clipped before deployment.
Open-loop performance	The plant drifts from the horizontal equilibrium and the cart model shows friction-dependent mismatch in step and frequency tests.	Hardware validation must report measured time/frequency behavior rather than relying only on the linear model.

Table 3: Open-loop analysis used before controller design.

3.7. Seesaw Parameter Estimation

While the manufacturer datasheet provides reliable nominal data for the majority of the seesaw parameters, the rotational dynamics are highly sensitive to the exact physical parameter D_C (Distance from the pivot to the Center of Gravity). To estimate this value, a data-driven Quasi-Static Lift-off Test was executed.

3.7.1. Quasi-Static Methodology

The physical procedure and software extraction pipeline involves:

1. **Immobilizing the module:** The seesaw is laid against the table at a known cart resting position $x_c(0) \simeq -0.407$ [m] and resting angle $\theta_{rest} \simeq \theta_{range} = 11.5^\circ$.
2. **Quasi-Static Actuation:** A constant voltage is applied to move the cart toward the pivot, effectively ensuring that the velocity remains negligible ($\dot{x}_c \approx 0, \dot{\theta} \approx 0$), leaving only the gravitational and static torque balances active.
3. **Lift-Off Detection:** The data-processing script scans the telemetry array to isolate the exact index where the angle breaks from θ_{rest} by a kinematic threshold corresponding to 3 encoder ticks ($3 \cdot K_{E_SW}/K_{sg} = 0.0015$ rad).

The cart's coordinate at this threshold is recorded as x_{lift} .

At the moment of lift-off, the static torques balance. The sum of moments about the pivot yields:

$$gM_{tot} \cdot x_{lift} = g(M_{tot}D_T + M_{SW}D_C) \cdot \theta_{rest} \quad (26)$$

Rearranging this equilibrium equation isolates the true center of gravity offset:

$$D_C = \left| \frac{M_{tot} \cdot x_{lift} - M_{tot} D_T \cdot \theta_{rest}}{M_{SW} \cdot \theta_{rest}} \right| \quad (27)$$

This calculation overwrites the nominal D_C value, directly refining the A_{SW} and B_{SW} matrices prior to feedback controller synthesis.

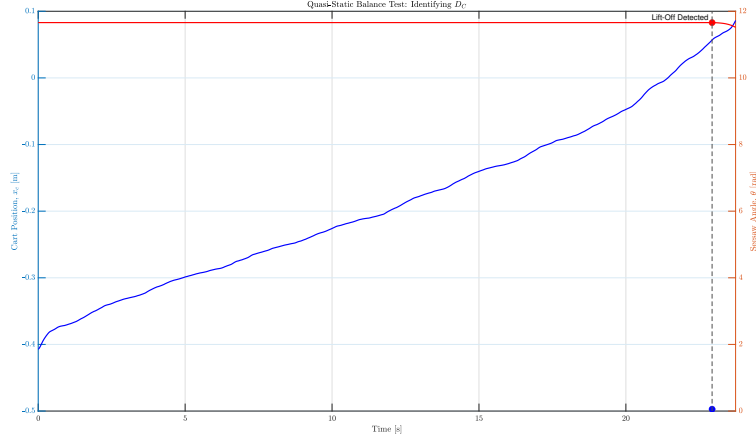


Figure 4: Seesaw Lift-Off Verification. The dual-axis plot illustrates the quasi-static progression of the cart (blue) and the resulting angular break (red). Circular markers designate the detected lift-off point used to compute D_C .

Executing this quasi-static procedure on the hardware and applying the lift-off torque balance yields the refined center-of-gravity offset:

$$D_C = \left| \frac{0.75 \cdot 0.0568 - 0.75 \cdot 0.125 \cdot 0.20}{3.6 \cdot 0.20} \right| = 0.033 \quad (28)$$

3.8. Closed-Loop Design Specifications

Before comparing controllers, the self-imposed closed-loop specifications were fixed around the hardware limits and the open-loop instability. Table 4 collects the targets that are otherwise used throughout the controller sections.

Task	Specification	Reason
Safety envelope	Keep $ V_m $ within the deployed clamp, avoid track end-stops, and limit high-frequency rack force.	Protect the amplifier, gearbox, and cart travel.
Cart inner loop	Tune a PID loop with phase margin $\geq 65^\circ$; retune to $\omega_{c,in} \approx 11.1$ rad/s and PM $\approx 85^\circ$ for the cascade.	Keep the cart loop faster than the angle loop but avoid noise amplification.
Cascaded angle loop	Use negative-gain PD/PID synthesis around $\omega_n = 4.3$ rad/s and $\zeta = 0.6$; add integral action for bias rejection.	Stabilize the RHP pole and remove steady-state tilt under constant load.
Pole placement	Dominant pair $-5.50 \pm j4.125$, non-dominant poles $-8.25, -6.60$, integral pole -0.5 .	Obtain a faster but voltage-limited state-space benchmark.
LQR/LQI	Bryson limits from <code>lqr_design.m</code> : $ x_c \leq 0.05$ m, $ \theta \leq 1^\circ$, $ \dot{x}_c \leq 0.1$ m/s, $ \xi \leq 4^\circ$ s, $ V_m \leq 3$ V.	Trade tracking, cart travel, integral recovery, and voltage effort systematically.
Validation	For each angle controller: simulate design, deploy common $\pm 1^\circ$ step plus stepped-sine protocol, and report time/frequency metrics.	Make the final comparison fair and tied to hardware evidence.

Table 4: Self-given closed-loop specifications used for the controller designs.

4. Frequency-Based Control: Cart Position Control Design

This section uses classical frequency-domain techniques. The objective is to design a feedback controller $C_x(s)$ that shapes the open-loop transfer function $L_x(s) = C_x(s)G_x(s)$ to satisfy specific stability margins and crossover frequencies. This cart position controller then serves as the high-bandwidth inner loop within a cascaded controller for stabilizing the seesaw angle.

4.1. Feedback Controller Tuning

The inner-loop PID was tuned by frequency-point shaping. The expected outer-loop bandwidth is constrained by the seesaw's RHP pole and RHP zero; taking the geometric mean of these bounds gives $\omega_{c,out} \approx 4.45$ rad/s. A five-times separation initially set $\omega_{c,in} \approx 22.25$ rad/s, while a 65° phase-margin target limited derivative noise and overshoot.

The selected controller is a PID with a low-pass filter (N-filter) on the derivative term to ensure the transfer function remains strictly proper. The control law is formulated as:

$$C_x(s) = K_{p,in} + \frac{K_{i,in}}{s} + \frac{K_{d,in} \cdot N_{in} \cdot s}{s + N_{in}} \quad (29)$$

The gains are obtained by evaluating $G_x(j\omega_{c,in})$, imposing $|C_x G_x| = 1$ and the desired phase at crossover, and solving the resulting real/imaginary equations for K_p and K_d while setting $K_i = K_p/T_i$. In compact form, the target controller vector is $C_{target} = |G_x(j\omega_{c,in})|^{-1} e^{j\phi_c}$, where $\phi_c = -180^\circ + \Phi_{PM,in} - \angle G_x(j\omega_{c,in})$.

4.1.1. Simulation Validation and Constraints Check

Because the physical hardware is subject to mechanical stress and current limitations, the controller was validated in simulation prior to deployment. To reduce

control aggressiveness and limit the initial current surge during step transitions, a second-order Bessel pre-filter $F_{bessel}(s)$ with a cutoff frequency $\omega_{f,bs} = 3.5$ rad/s is applied to shape the position reference signal:

$$F_{bessel}(s) = \frac{3 \cdot \omega_{f,bs}^2}{s^2 + 3 \cdot \omega_{f,bs} \cdot s + 3 \cdot \omega_{f,bs}^2} \quad (30)$$

The simulation includes friction compensation and checks the same voltage, current, track, and rack-force limits used in the open-loop validation.

4.2. Time Validation

Following mathematical verification, the inner-loop tracking performance was deployed on the physical testbed.

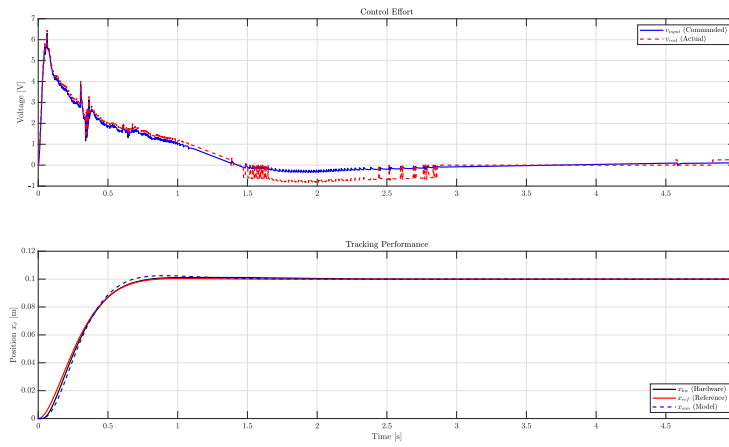
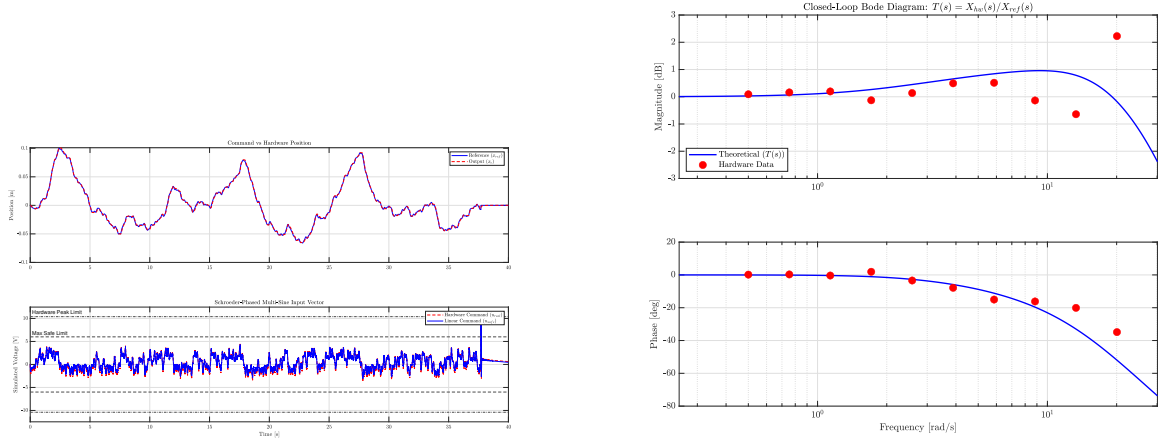


Figure 5: Filtered Step Response of the Cart Position Controller. Comparison between the predicted physics simulation (incorporating friction compensation) and the actual hardware telemetry.

The cart reaches the neighbourhood of the reference in roughly 1.2 s, after which the integral action converges more slowly to the exact position. The hardware trace also exposes a side effect of the friction compensation: with the deadzone threshold set conservatively low ($\epsilon = 0.1$), the inverse-deadband injection slightly degrades the response rather than improving it, indicating that the chosen threshold sits below the level actually required to break stiction.

4.3. Frequency Validation

To validate the inner-loop bandwidth, a closed-loop Schroeder multi-sine position reference was executed from 0.5 rad/s up to a force-limited maximum. Amplitude tapering and smooth fade-in kept high-frequency rack forces within the safe range. Zooming in on the position trace, the hardware oscillations are noticeably more attenuated than the simulation predicts, while the input voltage is slightly smoother than the simulated command even though its overall shape is preserved.



(a) Closed-loop Schroeder-phased multi-sine profile. Top: (b) Closed-loop cart position Bode validation. The empirical reference trajectory tracking and hardware response. Bottom: hardware data points are compared with the analytical model.

Figure 6: Closed-loop cart position controller validation in the time and frequency domains.

As anticipated from the multi-sine profile, the measured closed-loop magnitude rolls off earlier than the model predicts: across the cart step, the multi-sine, and this Bode estimate, the hardware consistently behaves as a lower-bandwidth, better-damped system than the nominal simulation. This is attributed to unmodelled friction that the inverse-deadband compensation cannot fully cancel, which both attenuates the high-frequency oscillations and limits the effective tracking bandwidth. The single magnitude outlier is most likely a numerical-conditioning artefact at that frequency rather than a physical resonance.

5. Frequency-Based Control: Seesaw Angle Control Design

5.1. Inner-Loop Re-Tuning

When the cart loop was embedded in the seesaw cascade, the initial bandwidth was too aggressive for the coupled hardware: the measured cart loop already rolled off earlier than the model, and the outer loop amplified the remaining derivative noise into unnecessary voltage motion. The deployed cascade therefore used a retuned inner loop with $\omega_{c,in} \approx 11.1$ rad/s, phase margin near 85° , derivative filter coefficient $N_{in} = 100$, a slower integrator ratio. This is a deliberate robustness trade-off: the cart remains faster than the angle loop, but the inner loop no longer tries to force high-frequency behaviour that the real track cannot reproduce.

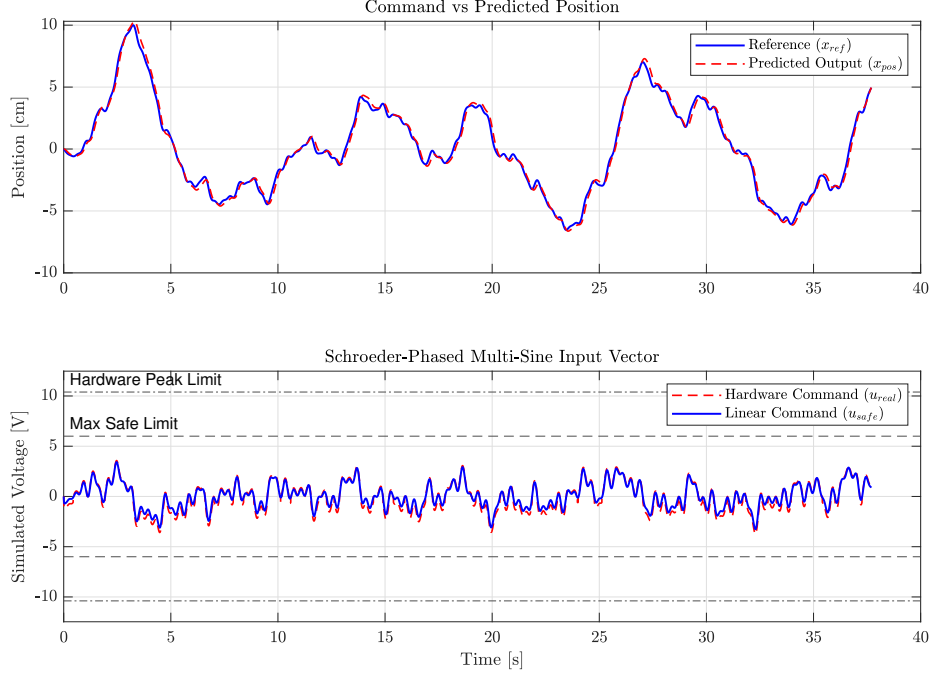


Figure 7: Simulation check for the retuned inner cart loop before closing the outer seesaw loop.

5.2. Cascade Architecture and System Limitations

To satisfy both stability and performance criteria for the highly coupled cart-seesaw system, a cascade controller is used. The physical dynamics are decoupled into two hierarchical loops: a high-bandwidth inner loop tracking cart position (x_c), and a slower outer loop dictating the required cart position to stabilize the seesaw angle (θ).

The effective plant visible to the outer-loop controller, denoted $P_{out}(s)$, is the series combination of the closed inner-loop transfer function $T_{in}(s)$ and the cart-to-angle cross-coupling dynamics $G_\theta(s)$:

$$P_{out}(s) = T_{in}(s)G_\theta(s) \quad (31)$$

As explained during the transfer function definition (sec. 3.4), the plant impose conflicting theoretical bounds on the achievable closed-loop bandwidth, necessitating highly precise dominant-pole placement to outrun the unstable exponential drift caused by RHP pole without exacerbating the RHP zero's initial undershoot.

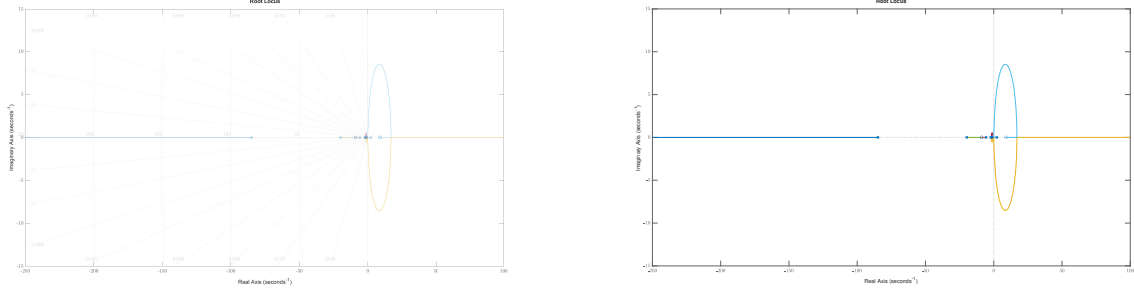
5.3. Outer-Loop Feedback Controller Tuning

Because heuristic PID tuning is insufficient for stabilizing this NMP system, we used phase-deficit design. The objective is to compute exact gains for a Lead Compensator (PD) and a PI-Lead Compensator (PID) to place the dominant closed-loop pole pair at a natural frequency $\omega_n = 4.3$ rad/s with a damping ratio $\zeta = 0.6$:

$$s_d = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} \quad (32)$$

A standard root locus analysis of $P_{out}(s)$ indicates that the RHP pole and RHP zero heavily attract one another: applying a positive proportional gain immedi-

ately drives the closed-loop poles deeper into the RHP. Consequently, a negative overall gain is used, which shifts the root-locus paths by 180° and moves the RHP pole into the Left-Half Plane (LHP), as illustrated below.



(a) Zoomed close to the origin.

(b) Zoomed out to show asymptotic paths.

Figure 8: Root-Locus of the outer-loop plant $\text{rlocus}(-P_{out})$.

5.3.1. PD (Lead) Synthesis

For the baseline proportional-derivative approach, a lead compensator structure is defined with a fixed derivative filter pole positioned at $p_{lead} = 25 \text{ rad/s}$.

$$C_{\theta,PD}(s) = -K_{pos} \frac{s + z_{lead}}{s + p_{lead}} \quad (33)$$

The required lead zero (z_{lead}) is computed by evaluating the phase deficit of the uncompensated plant exactly at the target pole s_d . The magnitude condition at s_d subsequently yields the necessary stabilizing negative gain ($-K_{pos}$).

5.3.2. Constant Bias Torque and Integral Action Requirement

While the PD compensator successfully stabilizes the seesaw around the horizontal equilibrium during standard transient events, it inherently lacks the low-frequency gain required to reject constant input disturbances. If an asymmetric load, such as a 100 g offset mass, is applied to the module, the proportional effort is eventually balanced by the disturbance torque, forcing the system to settle at a non-zero steady-state angular tilt.

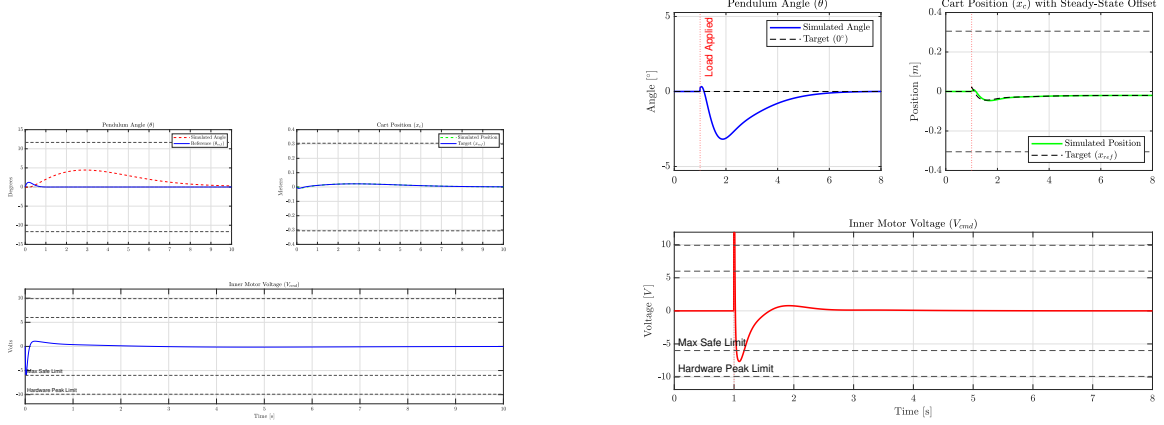
To remove this residual error, the controller is upgraded to a PI-Lead (PID) topology by introducing integral action. An integrator zero is placed exactly a decade below the derivative pole ($z_i = 2.5 \text{ rad/s}$) to preserve the transient shape established by the Lead compensator.

$$C_{\theta,PID}(s) = -K_{pos} \frac{(s + z_{lead})(s + z_i)}{s(s + p_{lead})} \quad (34)$$

This integrator accumulates the steady-state error, ramping the commanded cart displacement until the disturbance is canceled and the seesaw returns to the 0° equilibrium.

The constant load bias is modeled as an equivalent offset at the plant input, corresponding to a 2 cm step in the commanded cart position.

The simulated response spikes at the instant the bias is applied; this transient is an artifact of the idealized step in commanded position and would not appear on the hardware. Still, it motivates adding a safety saturation block on the voltage command prior to deployment.



(a) PD lead response to a 1° disturbance.

(b) PID rejection of the equivalent 100 g bias load.

Figure 9: Compact simulation checks for the cascaded angle controller: stabilization by lead action and bias rejection by integral action.

5.3.3. Architecture comparison

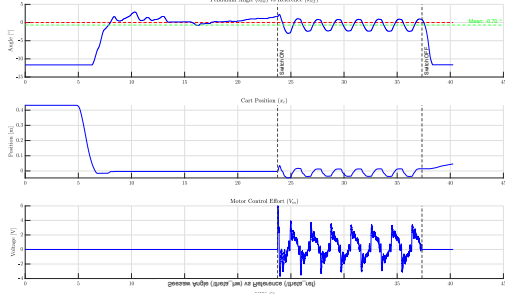
The integral controller removes the simulated steady-state bias left by the PD structure, but it also increases low-frequency loop authority. The hardware validation below therefore treats the PID as the deployable angle-tracking controller and keeps the PD mainly as a regulation baseline.

$$\begin{aligned}
 \text{pole}(P_{out}) &= [-84.77; -19.79; -5.58; \\
 &\quad 2.21; -2.21; -1.50] \\
 \text{pole}(T_{out,PD}) &= [-89.37; -20.14 + 5.34i; -20.14 - 5.34i; -5.71; \\
 &\quad -0.48 + 0.55i; -0.48 - 0.55i; -0.33] \\
 \text{pole}(T_{out,PID}) &= [-105.96; -10.39 + 10.52i; -10.39 - 10.52i; -2.58 + 3.44i; \\
 &\quad -2.58 - 3.44i; -3.04; -0.85 + 0.40i; -0.85 - 0.40i]
 \end{aligned} \tag{35}$$

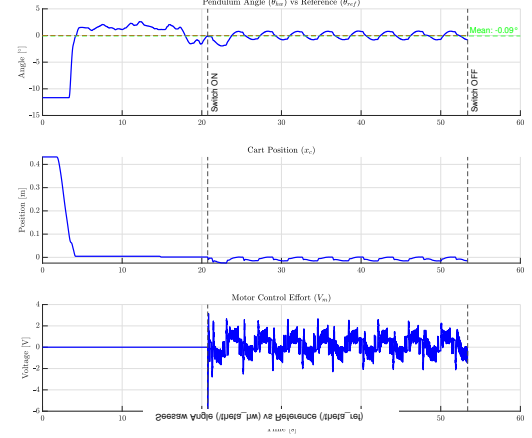
Both the PD and PID designs move the unstable right-half-plane pole into the left-half plane, confirming closed-loop stabilization in the linear model before hardware deployment.

5.4. Time Validation: Free-runs & Step-Response

The first hardware check was a zero-reference free-run. With only outer PD action, the seesaw remains stable but rocks around a biased equilibrium: RMS angle is about 3.5°, with many samples close to the mechanical limit. Adding the integral branch reduces the residual rocking to 0.69° RMS and shifts the mean angle from about -0.70° to -0.09°. A separate 1° PID step peaks near 3.7° and settles slowly, motivating the common validation protocol below.

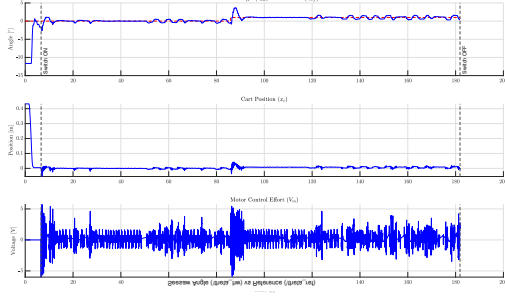


(a) PD zero-reference free-run.

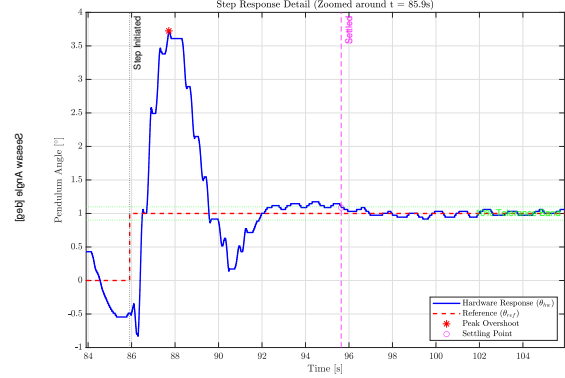


(b) PID zero-reference free-run.

Figure 10: Cascaded seesaw hardware validation: zero-reference free-run regulation comparing PD and PID controllers.



(a) PID 1° step full profile.



(b) PID 1° step zoom.

Figure 11: Cascaded seesaw hardware validation: dedicated PID 1° step response showcasing the full profile and zoomed-in settling behavior.

The numerical summary in Table 5 shows why the integral branch was kept for the deployed cascade. The PD controller stabilizes the plant, but the mean angle and spread show a persistent bias. The PID branch removes most of that bias, although the dedicated step run confirms that this classical cascade remains strongly underdamped on the real mechanism.

PD/PID Hardware Time Runs Comparison

Metric	PD	PID
Mean angle [°]	-0.70	-0.09
Min angle [°]	-2.98	-1.95
Max angle [°]	2.23	0.83
Cycle time [s]	1.90	3.57

PID step metric	Value
Step size [°]	1.0
Peak value [°]	3.72
Overshoot [%]	272.4
Settling time [s]	9.73
Steady-state error [°]	-0.01

Table 5: Dedicated free-run and step-response metrics before the shared tracking protocol.

5.5. Frequency Validation: Reference Tracking Protocol

The final cascaded PID was then evaluated on the same reference-tracking protocol later used for pole placement and LQR. The plotted results and Table 8 come from the large hardware validation run set: the same extraction script trims level segments, computes step metrics, estimates the closed-loop response at each sine segment, and reports the regulation spread and actuator effort. The run combines $\pm 1^\circ$ steps and a short pulse with a stepped sine sweep from 0.10 to 10 Hz, giving both a transient comparison and a small-signal frequency-response estimate without injecting an unsafe broadband command into the hardware.

Rather than using a multi-sine approach, the protocol automatically injects a continuously concatenated sequence of analytical commands, $r_\theta(t)$:

1. **Baseline & Step Analysis:** Initial static regulation followed by $\pm 1.0^\circ$ step commands and a 0.3-second, 1.0° impulse to characterize transient settling times and integral windup rejection.
2. **Stepped Sine Sweep:** A series of logarithmically spaced sinusoidal tracking commands spanning from 0.10 Hz to 10.00 Hz.

To ensure hardware survivability during the high-frequency sine sweeps, two safeguards were embedded into the reference trajectory:

- **Dynamic Amplitude Tapering:** Below 2.5 rad/s, the reference amplitude is held at 0.65° to reliably overcome static friction. Above this threshold, the amplitude is aggressively tapered proportional to $1/\omega^2$. This keeps the acceleration forces acting on the cart rack constant and within the safe limit, despite the increasing frequency.
- **Smooth Cyclic Enveloping:** Each frequency segment runs for exactly 7 cycles. The first cycle is multiplied by a half-cosine ramp-up envelope, and the final cycle uses a half-cosine ramp-down envelope. This prevents discontinuous velocity shocks at the boundary transitions between differing frequencies.

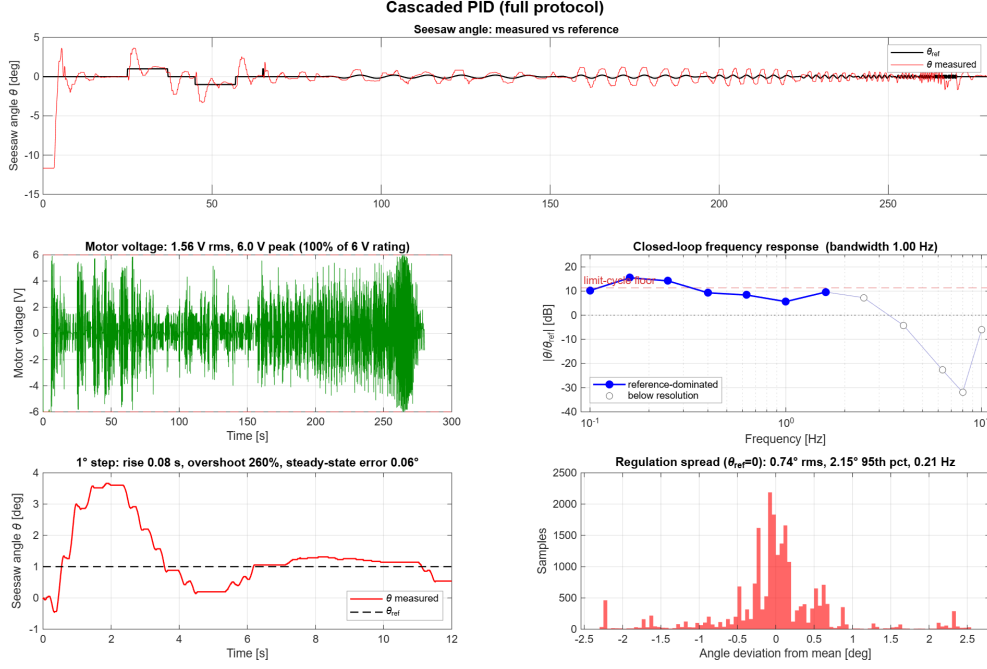


Figure 12: Cascaded PID, full tracking protocol: angle reference and measurement, motor voltage, closed-loop response estimate, $+1^\circ$ step detail, and the regulation spread at $\theta_{ref} = 0$.

The PID tracks the $+1^\circ$ step with a steady-state gain close to unity and only 0.06° residual error, but it is strongly underdamped: the measured angle overshoots to roughly 3.6° before returning to its rocking band. Across the sine segments, the RMS tracking error is 0.52° . It also uses the most actuator authority of the three angle controllers, with 1.56 V RMS motor effort and 11.6 cm peak-to-peak cart travel.

6. State-Space Control Design: Pole Placement

6.1. Full-State Feedback

The state-space controller uses the full-state feedback law $V_m = -Kx$ with the continuous-time 4-state plant derived in the modeling section. The primary control objective is to regulate the seesaw angle such that $\theta \rightarrow 0$, with the cart's translation serving to maintain this angular balance. For hardware implementation, only the cart position x_c and the seesaw angle θ are directly measured via encoders; the velocity states $\dot{x}_c$ and $\dot{\theta}$ are estimated using an observer.

6.1.1. Baseline Stabilizing Design

The initial stabilization attempt (Attempt 1) establishes a baseline controller by mirroring the unstable pole directly into the LHP while maintaining the remaining poles at their original open-loop locations.

The desired pole locations and resulting feedback gain vector are:

$$\begin{aligned} P_{des} &= \{-0.42, -2.25, -2.61, -36.50\} \\ K_1 &= [78.36, -59.81, 4.46, -22.90] \end{aligned} \quad (36)$$

While this baseline design successfully stabilizes the system in simulation, the transient response is excessively sluggish.

6.1.2. Limit-Cycle Minimization & Dominant-Pole Redesign

To address the sluggish transient response and minimize the steady-state limit cycle caused by hardware non-linearities (Coulomb friction, encoder quantization, gear backlash), the controller was redesigned. The closed-loop poles were strategically placed to enforce a specific frequency-domain separation between the cart and the seesaw dynamics.

To outrun the divergence while preventing control efforts that would saturate the 6 V actuator limit, the design enforces a strict hierarchy: the complex conjugate pair governing the seesaw angle (θ) acts as the dominant mode, while the real poles governing the cart are placed at higher frequencies to maintain a $\approx 2 : 1$ bandwidth window.

The selection parameters are:

- **Dominant Angular Pair ($p_{1,2}$):** A damping ratio of $\zeta_\theta = 0.8$ was selected to balance settling time and stability margins. The real part (σ_θ) dictates the error correction rate and was set to 5.5 rad/s to confidently outpace the open-loop instability.
- **Cart Velocity Pole (p_3):** Placed at $1.5 \cdot \sigma_\theta$ so that the cart velocity settles faster than the balancing mode.
- **Cart Position Pole (p_4):** Placed at $1.2 \cdot \sigma_\theta$ to maintain the dominance of the angular pair while setting a reliable return-to-center time constant.

This logic yields the following analytical pole placements:

$$\begin{aligned} p_{1,2} &= -5.50 \pm j4.125 \\ p_3 &= -8.25 \\ p_4 &= -6.60 \end{aligned} \quad (37)$$

Pole-Placement Evolution

Parameter	Attempt 1	Final Design	Unit
Poles	$[-0.42, -2.25, -2.61, -36.50]$	$[-5.50 \pm j4.13, -8.25, -6.60]$	rad/s
Peak V_m (Sim)	4.70	≈ 6.00	V
Settling Time	> 10	< 2.0	s

Table 6: Pole-placement evolution on the tuned plant. The final design isolates and accelerates the dominant dynamics.

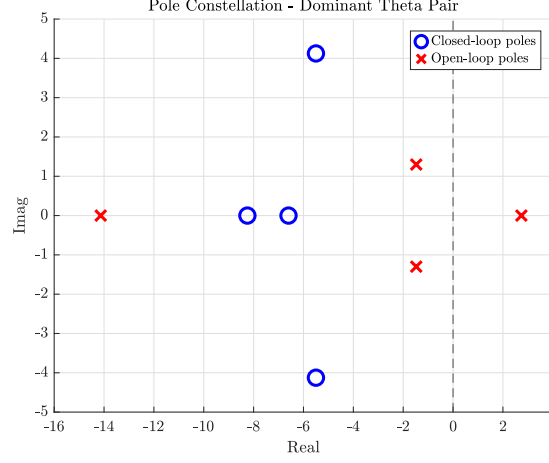
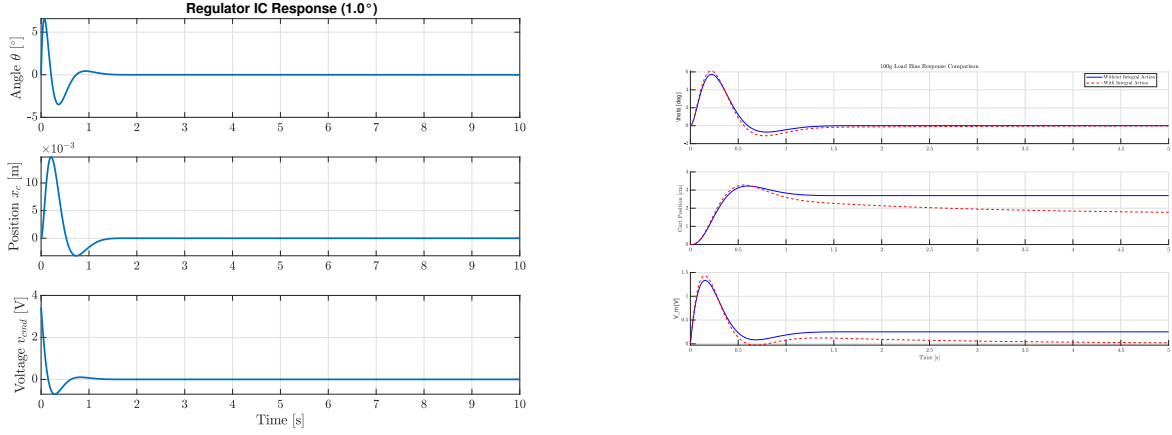


Figure 13: Pole constellation comparing open-loop plant instability to the target closed-loop dominant theta pair.



(a) Final pole-placement response to a 1° initial tilt. (b) Bias-load comparison with and without integral action.

Figure 14: Pole-placement simulation checks before hardware validation.

6.1.3. Constant Bias Torque and Integral Action

While proportional state-feedback ensures transient stability, it fails to eliminate steady-state error under persistent asymmetric loads. During a simulated bias test representing a 100 g off-center mass, the system slightly shifts its oscillation center away from $\theta = 0^\circ$, exhibiting a constant residual tilt.

To eliminate this steady-state tracking error, the controller uses the augmented 5-state model (A_5, B_5) introduced in the state-space derivation, with the integral of the angle error ($\xi = \int \theta dt$).

To preserve the fine-tuned transient shape established by the dominant pair, the four existing poles are retained, and the new integral pole (p_{int}) is placed conservatively at a slower frequency of -0.5 rad/s. The simulation confirms that the augmented integral controller drives the angular error to zero, stabilizing the cart at an offset positional equilibrium.

In the deployed controller this augmented gain is the version used for reference

tracking. The pure four-state gain is still useful as a regulation baseline, but the integral state is required whenever a nonzero angle command or persistent bias has to be followed without steady-state error.

6.2. Time Validation: Horizontal Stability

Prior to executing the tracking protocol, a free-run test of the standard pole-placement controller was conducted on the physical hardware. With the reference held at 0° , the controller holds the body in a tight bounded rocking band of 0.32° RMS, the smallest regulation error among the three tested angle controllers. The motor effort is also much lower than the cascaded PID, at approximately 0.56 V RMS. The remaining motion is best interpreted as a friction/quantization limit cycle rather than a failure to stabilize: the open-loop unstable plant is held near the horizontal equilibrium and does not drift.

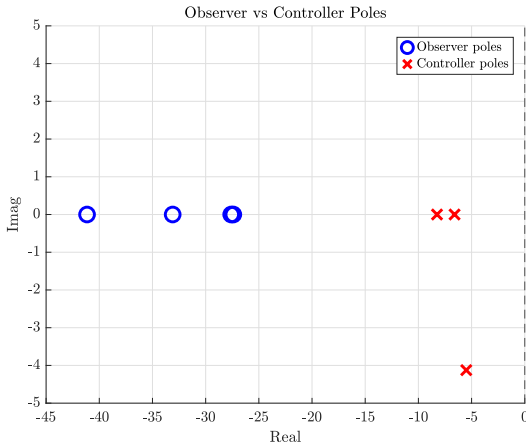
6.3. State Estimator: Luenberger Observer

Because raw numerical differentiation of the encoder ticks heavily amplifies quantization noise, a full-order Luenberger observer is implemented to estimate the unmeasured velocity states robustly.

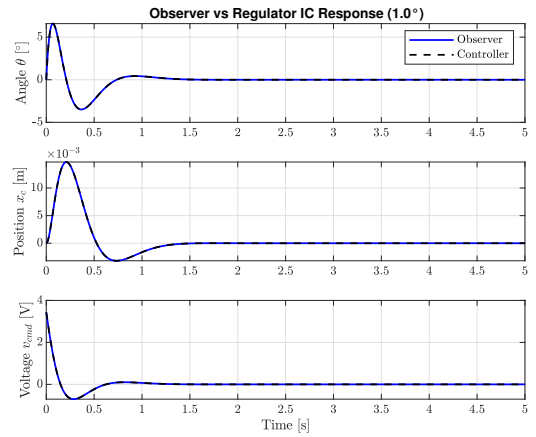
The observer gain matrix L is computed by assigning the estimator poles to be strictly faster than the controller poles. A proportional scaling factor of $k_{obs} = 5.0$ is applied to the real components of the desired closed-loop poles.

To prevent the ‘place’ algorithm from failing due to exact pole multiplicities (which degrade numerical computation), a microscopic separation delta ($\delta = 0.1$) is systematically applied to the target real poles:

$$p_{obs} = \begin{bmatrix} 5.0 \cdot \text{Re}(p_{des,1}) + \delta \\ 5.0 \cdot \text{Re}(p_{des,2}) - \delta \\ 5.0 \cdot p_{des,3} + \delta \\ 5.0 \cdot p_{des,4} - \delta \end{bmatrix} \quad (38)$$



(a) Controller and observer pole locations.



(b) Simulated observer convergence from a 1° initial tilt.

Figure 15: Pole-placement observer design checks before hardware logging.

On hardware, however, the observer was logged in parallel while the feedback loop still used filtered encoder derivatives. The logged comparison is therefore an estimator validation, not an in-loop observer-control validation. The angle innovation is small, about 0.009° RMS, and the observer’s angular-rate estimate is smoother than the dirty derivative at high frequency, where encoder quantization is most visible. This supports using a model-based estimator in the next iteration, but the observer bandwidth would need to be reduced before closing the loop through it because the aggressive $k_{obs} = 5$ design would amplify angle encoder ticks into voltage chatter.

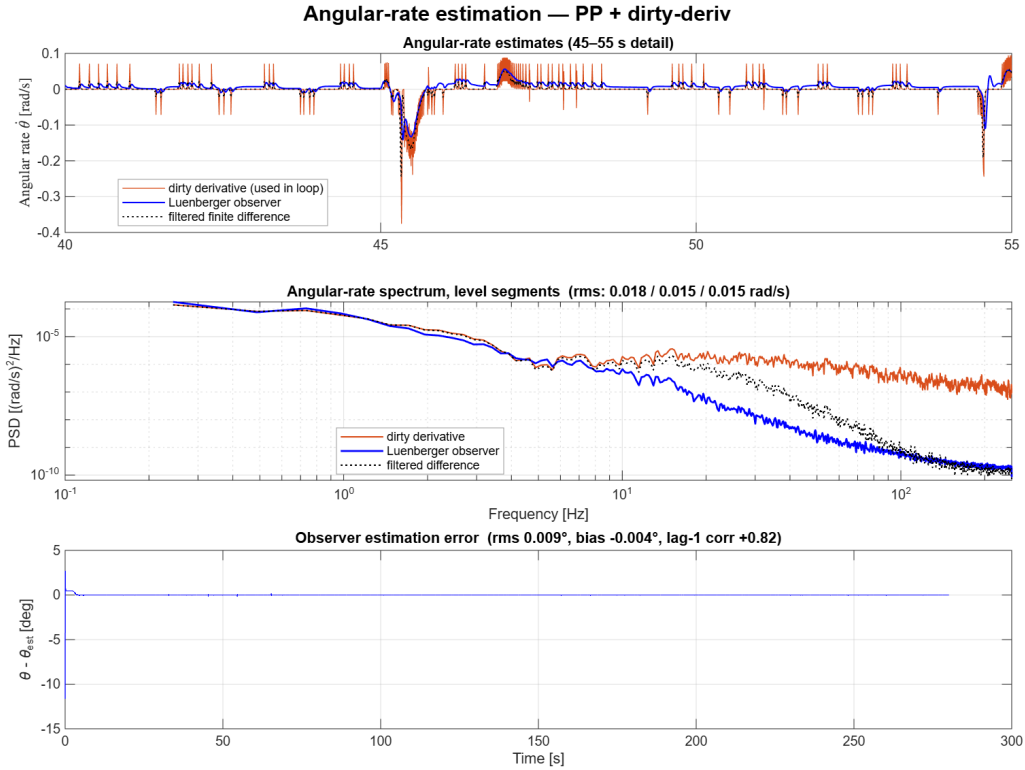


Figure 16: Pole-placement run: dirty-derivative, observer, and filtered finite-difference angular-rate estimates, with spectra over the level segments and the observer innovation.

6.4. Digital Implementation and Frequency Validation

Prior to deployment on the real-time hardware, the continuous-time matrices and filters are discretized to match the hardware sampling time T_s . The plant state-space is converted via Zero-Order Hold (ZOH), while auxiliary derivative filters are mapped using the Tustin (Bilinear) approximation to preserve high-frequency phase fidelity.

An analytical loop analysis evaluated the robustness of the dominant-pole design. Broken at the plant input, the deployed loop has a phase margin of 58.4° and a crossover frequency of 12.9 rad/s. On the real-time target, only x_c and θ are measured directly. The implementation therefore used dirty-derivative estimates for

$\dot{x}_c$ and $\dot{\theta}$, with first-order low-pass filtering to limit encoder noise. The reference-tracking run used the integral-augmented gain and the same protocol introduced in the PID section.

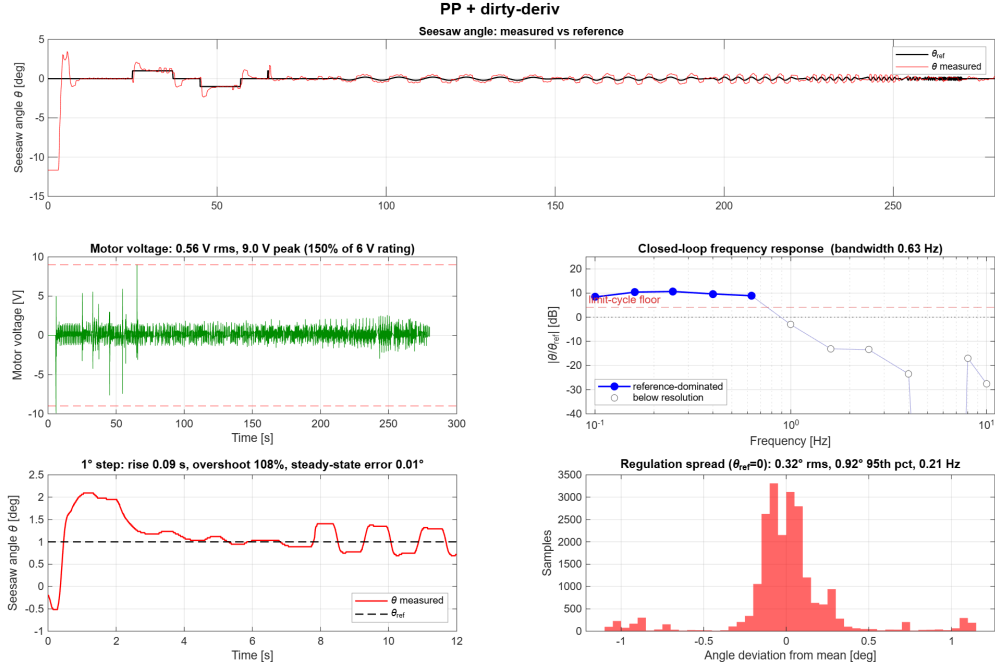


Figure 17: Pole placement with dirty-derivative velocities, full tracking protocol: angle reference and measurement, motor voltage, closed-loop response estimate, $+1^\circ$ step detail, and regulation spread.

With integral action active, the pole-placement controller tracks the $+1^\circ$ step with gain 1.01 and only 0.01° steady-state error. Its peak response is about 2.1° , making it the best-damped of the three measured controllers. Across the stepped sines it gives the lowest RMS tracking error, 0.33° , while keeping the motor effort near 0.56 V RMS and cart travel to 7.5 cm peak-to-peak. The state-space runs used a higher voltage clamp than the cascaded PID, so the transient voltage peaks should be read against that implementation detail.

7. State-Space Control Design: Linear Quadratic Regulator

7.1. Optimal Full-State Feedback Synthesis (LQR)

While the pole-placement technique allows for explicit dynamic shaping, it relies on heuristic iterations to balance tracking against actuator saturation. To obtain stability margins and effort minimization from an explicit cost function, a Linear Quadratic Regulator (LQR) is implemented.

The LQR/LQI design follows `scripts/control/lqr_design.m`, using the same state ordering as the derived seesaw model, $[x_c, \theta, \dot{x}_c, \dot{\theta}]$, and the augmented integral state $\xi = \int \theta dt$.

The LQR algorithm computes the optimal full-state feedback gain matrix K by minimizing a quadratic cost function J , based on the continuous-time Algebraic Riccati Equation:

$$J = \int_0^\infty (x^T Q x + u^T R u) dt \quad (39)$$

The dynamic behavior of the system is explicitly tuned by balancing two user-defined weighting matrices: the state error penalty Q and the actuator effort penalty R .

7.1.1. Tuning via Bryson's Rule

Rather than assigning poles directly, the LQR/LQI design uses Bryson's-rule state and effort penalties. The selected limits penalize cart displacement, seesaw angle, cart speed, integral angle error, and motor voltage while leaving angular speed unpenalized directly.

The rule is defined as:

$$Q_{ii} = \frac{1}{\max(|x_{i,dev}|)^2}, \quad R = \frac{1}{\max(|V_{m,dev}|)^2} \quad (40)$$

The limits used in `lqr_design.m` that were already summarized in Tab. 4 are reminded in Tab. 7.

LQR/LQI Acceptable Deviations (Bryson's Rule)

Parameter	Description	Value	Unit
req_dist	Max Cart Displacement	0.05	m
req_angle	Max Seesaw Angle	1.0	°
req_speed	Max Cart Speed	0.1	m/s
req_angspeed	Max Angular Speed	$+\infty$	rad/s
req_integral	Max Integral Accumulation	4.0	° · s
req_volt	Target Max Voltage	3.0	V

Table 7: Design targets used in `lqr_design.m` to compute the LQR/LQI weighting matrices.

By leaving the angular speed unconstrained ($+\infty$), the optimizer applies no direct penalty to $\dot{\theta}$. This avoids over-penalizing transient angular-rate estimates obtained from noisy encoder derivatives.

Applying Bryson’s Rule yields

$$Q_4 = \text{diag}(400, 3282.8, 100, 0), \quad R_4 = \frac{1}{9} \quad (41)$$

The voltage penalty is not a hard actuator clamp; saturation is still enforced and checked during simulation and hardware validation.

7.2. Simulation Validation

To evaluate the transient behavior of the LQR before hardware deployment, two simulated stress tests were conducted.

First, the simulation was subjected to a 1° initial-tilt disturbance. The standard 4-state LQR arrests the angular velocity and returns the seesaw to 0° without excessive cart travel or voltage saturation, validating the chosen Bryson’s parameters for local regulation.

However, because the LQR fundamentally acts as a proportional state-feedback controller, it struggles against persistent unmodeled forces. When subjected to a simulated 100 g asymmetric bias load (equivalent to a 0.123 Nm continuous torque), the standard LQR successfully avoids total instability but settles at a persistent steady-state tilt error.

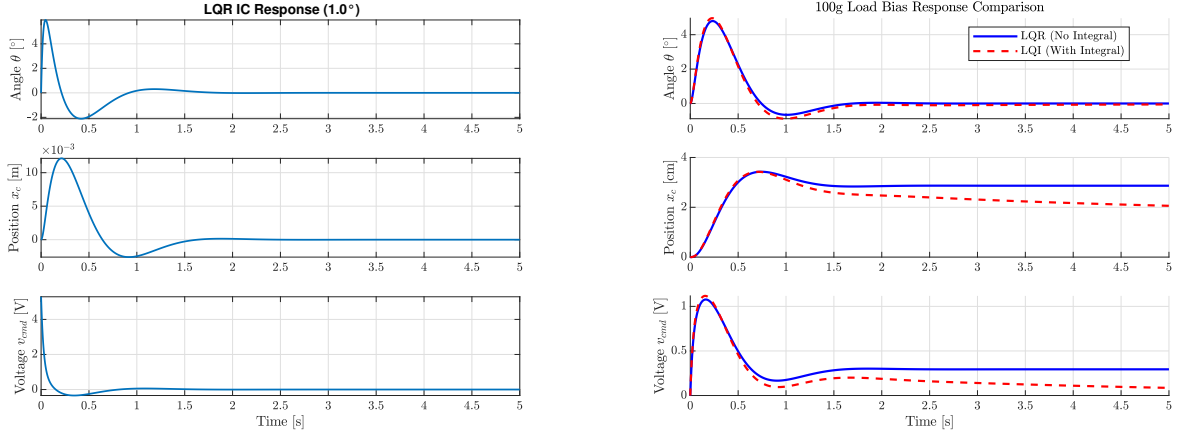
7.3. LQI: Integral Action for Disturbance Rejection

To remove the residual error identified during the simulation validation, the controller is upgraded to a Linear Quadratic Integral (LQI) controller using the augmented 5-state system ($\dot{\xi} = \theta$).

The weighting matrix Q_5 incorporates the original Bryson parameters while appending a penalty specifically for the integral state. The acceptable deviation for the integral was defined as $4.0^\circ \cdot \text{s}$, representing the maximum time-area allowed to clear a given angular deviation before heavy penalties apply:

$$Q_5 = \text{diag}(400, 3282.8, 100, 0, 205.2), \quad R_5 = \frac{1}{9} \quad (42)$$

Applying the 100 g bias load against this augmented LQI controller in simulation shows rejection of the constant disturbance, driving the steady-state angular error back to 0° .



(a) LQR response to a 1° initial tilt.

(b) LQR/LQI comparison under a 100 g bias load.

Figure 18: LQR/LQI simulation checks before hardware validation.

7.4. Time Validation: Horizontal Stability

Following theoretical and simulated validation, the LQI controller was deployed to the physical hardware using the same dirty-derivative velocity reconstruction as the pole-placement implementation. With the reference held at 0° , the loop holds the body to approximately 0.42° RMS. This is slightly looser than pole placement but substantially quieter than the cascaded PID, while requiring essentially the same RMS voltage as pole placement (0.58 V). A dedicated disturbance run also recovered to the same regulation band after manual perturbations, confirming that the controller stabilizes the plant robustly around the horizontal equilibrium.

7.5. State Estimator: Kalman Filter (LQE)

To estimate the unmeasured velocity states robustly, a Linear Quadratic Estimator (LQE), or Kalman Filter, was designed as a candidate replacement for the filtered derivative. The Kalman Filter computes the optimal observer gain L by treating the system stochastically, balancing expected modeling inaccuracies against known sensor precision.

The estimator uses two covariance matrices:

- **Process Noise Covariance (Q_n):** Represents the uncertainty in the mathematical plant model (e.g., unmodeled friction, backlash).
- **Measurement Noise Covariance (R_n):** Represents the electrical and quantization noise present in the physical sensors. This has been fixed to one encoder tick.

Because the incremental encoders provide high-fidelity positional measurements, while the dynamic velocity transitions are heavily subject to unmodeled frictional variations, the matrices were structured to mostly trust the sensors:

$$Q_n = \text{diag}(10^{-7}, 7.61 \times 10^{-7}, 10^{-6}, 10^{-6}), \quad R_n = \text{diag}(2.27 \times 10^{-5}, 1.5 \times 10^{-3}) \quad (43)$$

Because the ratio Q_n/R_n is structurally tuned, the Kalman filter converges on an optimal gain matrix L that minimizes the variance of the estimation error in the linear stochastic model.

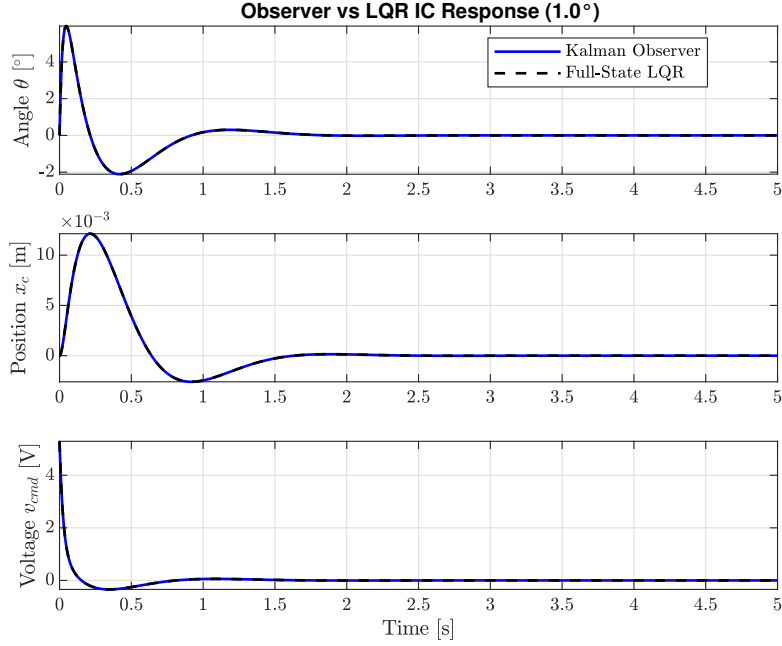


Figure 19: Simulated Kalman-filter convergence from a 1° initial tilt. This is the design-side convergence check.

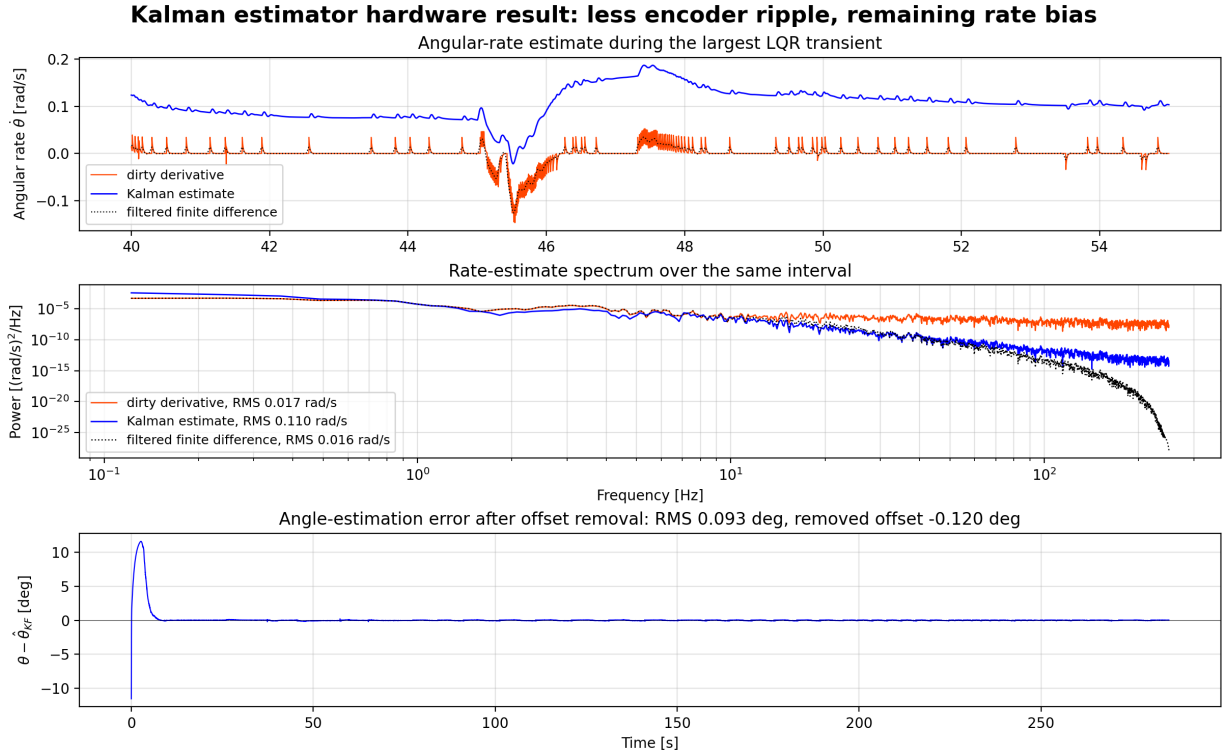


Figure 20: Hardware comparison of Kalman and dirty-derivative velocity estimates during the LQR test. The Kalman estimate reduces high-frequency encoder noise, but the residual low-frequency bias shows that the estimator still needs retuning before it is used for feedback.

The hardware comparison in Fig. 20 shows the main estimator trade-off. The Kalman angular-rate estimate removes much of the high-frequency encoder noise

that appears in the dirty derivative, giving a cleaner velocity signal for state feedback. However, the same run also shows a low-frequency bias in the estimated rate and a small residual angle-estimation error after the initial transient. This means the estimator improves noise rejection, but it is not yet clean enough to replace the dirty derivative in the feedback loop without further covariance retuning and a dedicated closed-loop Kalman test.

7.6. Digital Implementation and Frequency Validation

Similarly to Pole-Placement, the continuous-time matrices and filters are discretized to match the hardware sampling time T_s . In `lqr_design.m`, the continuous gain K_5 is computed with `lqr(A5,B5,Q5,R5)` and the discrete hardware gain K_{5d} with `dlqr(A5d,B5d,Q5,R5)`. The hardware was then run through the same reference protocol as the PID and pole-placement controllers, using dirty-derivative velocity reconstruction.

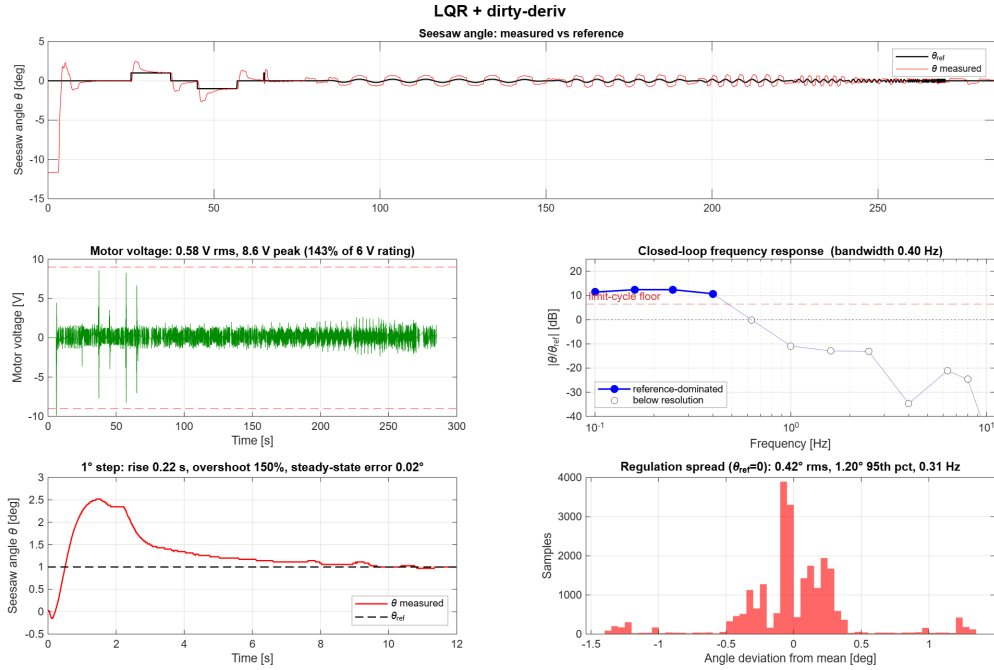


Figure 21: LQR/LQI with dirty-derivative velocities, full tracking protocol: angle reference and measurement, motor voltage, closed-loop response estimate, $+1^\circ$ step detail, and regulation spread.

The LQR tracks the $+1^\circ$ step with gain 1.02 and 0.02° steady-state error. Its rise time is the slowest of the three controllers, about 0.22 s, and the peak response reaches roughly 2.5° . Across the sine sweep, the RMS tracking error is 0.42° . The main practical advantage is low cart motion: the measured travel is about 5.0 cm peak-to-peak, the smallest of the tested controllers.

Large-angle tracking.

To probe how far the local controller could be pushed on hardware, the LQR was also driven through a staircase of increasingly large angle references up to 10° ,

close to the 11.5° physical stop. On the upward leg, the controller tracks each level with near-unity gain and no loss of stability, with peak voltage around 4 V. On the return, the measured angle sits roughly $1.5\text{--}2^\circ$ below the command at each level. This hysteresis is consistent with Coulomb friction and stiction, and it is one of the clearest hardware signatures of why the linear model overestimates damping.

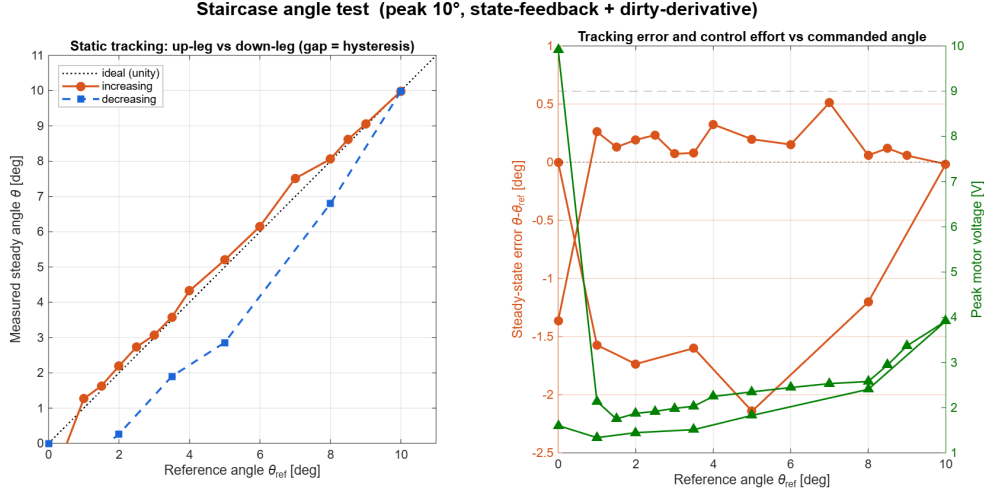


Figure 22: Large-angle LQR staircase test: static tracking remains close to unity up to 10° on the upward leg, while the return leg exposes a $1.5\text{--}2^\circ$ hysteresis loop.

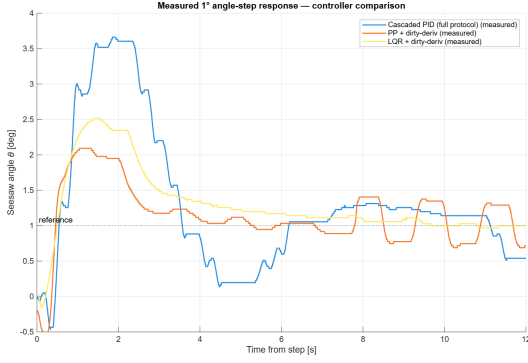
7.7. Controller Comparison and Design-versus-Reality

Table 8 summarizes the three closed-loop hardware controllers on the shared protocol. The cascaded PID is fastest but has the largest overshoot, rocking, motor effort, and cart travel. Pole placement gives the tightest small-signal tracking and regulation band. LQR is gentler on the cart and actuator while remaining close to pole placement in tracking accuracy.

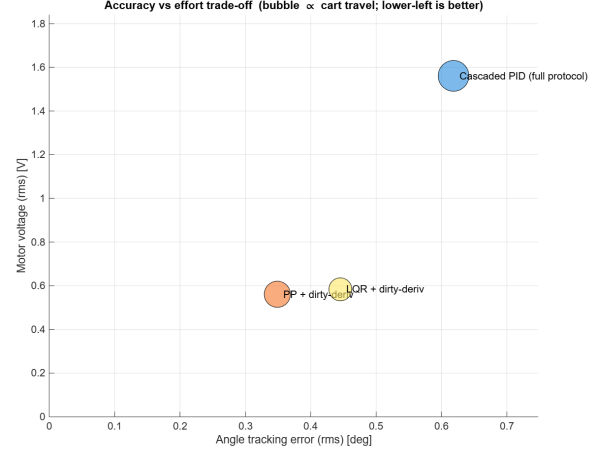
Measured Angle-Control Performance on the Common Tracking Protocol

Metric	Cascaded PID	PP+DD	LQR+DD
$+1^\circ$ step rise time [s]	0.08	0.09	0.22
$+1^\circ$ step peak [deg]	3.6	2.1	2.5
Steady-state step error [deg]	0.06	0.01	0.02
Swept-sine RMS error [deg]	0.52	0.33	0.42
Resonant peak $ \theta/\theta_{ref} $ [-]	6.0	3.4	4.2
Free-run rocking [deg RMS]	0.74	0.32	0.42
Motor voltage [V RMS]	1.56	0.56	0.58
Cart travel [cm p-p]	11.6	7.5	5.0

Table 8: Measured angle-control performance on the common tracking protocol.



(a) Measured 1° step responses.



(b) Accuracy vs. effort, bubble area proportional to cart travel.

Figure 23: Controller comparison on the identical hardware reference protocol.

The measured responses are less damped than the linear designs predict: every controller overshoots its 1° step and every swept-sine response shows a resonant peak. The common causes are implementation-level rather than controller-specific: dirty-derivative phase lag, Coulomb friction/stiction, backlash, and encoder quantization. The robust conclusions are the relative damping ranking, the near-zero steady-state errors from integral action, and the successful large-angle LQR staircase test.

8. Advanced Non-Linear Maneuver: Seesaw Lift-up

The lift-up problem is the hardest task because the motion starts from the mechanical stop, far from the horizontal equilibrium:

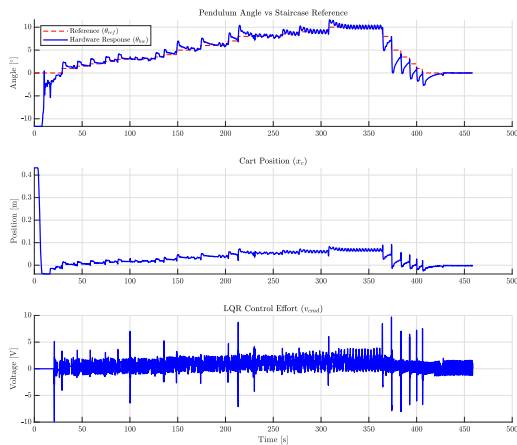
$$x(0) = [-0.407 \text{ m}, 11.66^\circ, 0, 0]^T \quad (44)$$

Because the previous linear controllers are valid only near $\theta = 0^\circ$, they cannot be engaged directly from this resting state without instability or actuator saturation. The lift-up design therefore estimates the local capture region and drives the plant into it before switching to the balancing controller.

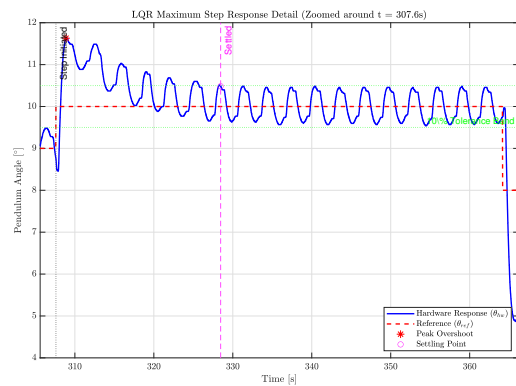
8.1. Region of Attraction and Capture Logic

The local controllers can only be trusted inside a bounded capture region around the horizontal equilibrium. This region was approximated with a quadratic Lyapunov level set $V(x) = x^T P x$, clipped by the same cart, angle, voltage, and slew-rate limits used elsewhere in the report. The large-angle staircase tests in Figs. 22 and 24 are used as the experimental counterpart to this calculation: the largest bounded offset gives the practical angle limit for slow entry into the local capture region. The tests support a 10° admissible offset, but they do not prove that a linear controller can lift the plant from the mechanical stop.

The matrix P comes directly from the Riccati solution for LQR/LQI; for pole placement, it is obtained from the continuous Lyapunov equation $A_{cl}^T P + P A_{cl} + Q = 0$. For each linear operating constraint $|Hx| \leq b$, the allowable Lyapunov level is bounded by $c = b^2 / (HP^{-1}H^T)$, and the capture threshold c^* is the minimum over the cart-position, angle, voltage, and slew-rate constraints. This gives a single switching rule rather than relying on a hand-tuned angle threshold alone.



(a) Full large-angle staircase run.



(b) Zoom around the largest step.

Figure 24: Time-domain staircase test used as the practical maximum-offset check. The response stays bounded for a 10° commanded offset, so this value is used as the experimentally supported boundary for slow entry into the local capture region.

8.2. Heuristic Lift-up via Gain Attenuation

Prior to formulating a fully optimized trajectory, an initial empirical lift-up maneuver was tested with an attenuated pole-placement controller to reduce the initial control effort.

Rather than engaging the full effort of the unaugmented Pole Placement controller from the resting state, the maneuver begins with attenuated feedback gains. By scaling the gains down to 30% of their nominal values, the controller applies enough control effort to induce a slow lift-off without causing actuator saturation or destabilizing the cart reference.

As the attenuated control effort swings the seesaw upward, the telemetry is continuously monitored. The exact moment the state vector crosses the mathematical boundary into the pre-computed Region of Attraction ($x^T P x < c^*$), the software immediately switches to the 100% nominal gain controller. Once inside the ROA, the standard controller safely arrests the residual momentum and locks the seesaw into horizontal equilibrium.

8.3. Theoretical Improvement: Trajectory Optimization & TV-LQR

The proposed lift-up improvement is a constrained CasADi trajectory optimization from $[-0.407 \text{ m}, 11.66^\circ, 0, 0]^T$ toward the horizontal equilibrium. The non-linear OCP uses $N = 100$ RK4 intervals and minimizes $J = w_T T + w_U \int_0^T U(t)^2 dt$ with $w_T = 1$ and $w_U = 0.1$, while enforcing $|x_c| \leq 0.432 \text{ m}$, $|V_m| \leq 6 \text{ V}$, and a feasible time window. This preserves the original objective: a short but physically survivable maneuver, not merely a simulated swing-up.

The feedforward path is stabilized by a TV-LQR schedule computed by integrating the Riccati equation backward from the terminal LQI cost. During the maneuver, the voltage is $V_m(t) = U_{tv}(t) - K_{tv}(t)(x(t) - x_{tv}(t))$, and a one-way switch hands control to the stationary LQI controller when $x_c > 0.057 \text{ m}$ and $|\theta| < 5^\circ$. Simulation verifies that the generated path respects the track and voltage limits, enters the Lyapunov capture region before handover, and avoids repeated switching.

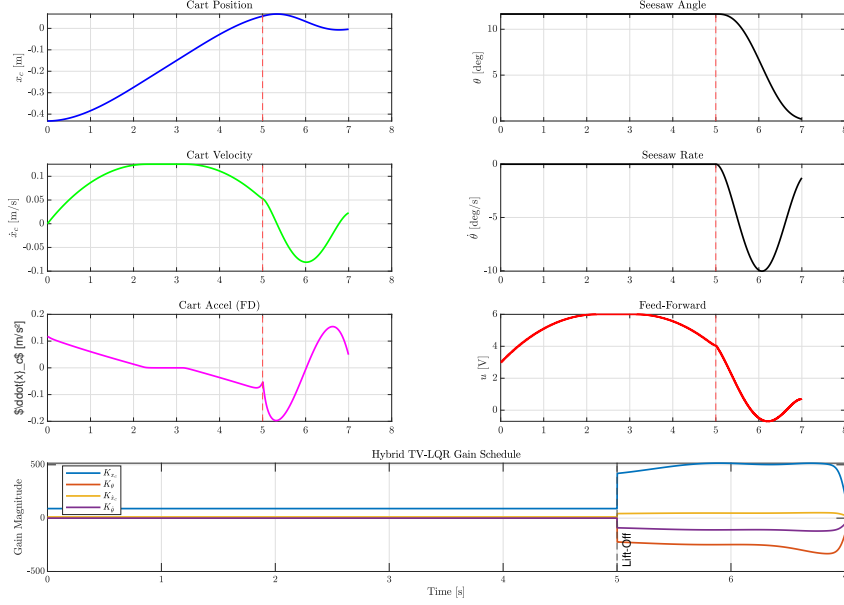


Figure 25: Optimized TV-LQR Lift-up maneuver. Displaying the feedforward voltage profile, state evolution, track limits, and the evaluation of the Lyapunov bound.

The validation checks are therefore fourfold: the cart stays inside the rail limits, the combined feedforward-plus-feedback voltage remains within ± 6 V, the Lyapunov value drops below the capture threshold before the stationary controller is armed, and the one-way `has_switched` flag prevents chatter at the handover boundary. These checks are simulation evidence only; the maneuver is proposed as the next hardware step rather than claimed as a completed deployment.

Optimal TV-LQR Lift-Up Parameters

Quantity	Symbol	Value	Unit
Initial state	x_{start}	$[-0.407, 11.66^\circ, 0, 0]$	-
Time penalty	w_T	1.0	-
Effort penalty	w_U	0.1	-
Cart limit	x_{max}	± 0.432	m
Voltage limit	V_{max}	± 6.0	V
LQI switch	x_c, θ	$> 0.057 \text{ m}, < 5^\circ$	-

Table 9: Numerical constants, weights, and constraints defining the CasADi trajectory optimization and TV-LQR handover.

9. Critical Analysis/Comparison of All Developed Control Strategies

Critical Comparison of Developed Control Strategies

Strategy	Main Objective	Strengths	Limitations	Final Role
Cart PID	Cart position control	Simple frequency-domain design; useful for validating the cart model, actuator behavior, bandwidth, and safety limits.	Sensitive to friction, derivative noise, and saturation; cannot stabilize the seesaw angle by itself.	Inner loop of the cascade controller.
Outer-loop PD / Lead	Horizontal seesaw stabilization	Stabilizes the unstable equilibrium through negative-gain lead compensation and root-locus design.	No integral action; constant bias torque produces steady-state angular error.	Baseline frequency-based angle controller.
Outer-loop PI-Lead / PID	Angle stabilization with bias rejection	Integral action removes steady-state error caused by constant disturbances, such as the 100 g asymmetric load.	More sensitive to windup and low-frequency oscillations; requires careful reference shaping and safety checks.	Improved frequency-based controller.
Pole Placement	State-space stabilization	Direct assignment of closed-loop poles; useful for shaping dominant dynamics and reducing settling time.	Model-dependent; does not directly optimize voltage effort; requires velocity estimation.	State-space benchmark controller.
Pole Placement with Integral Action	State-space disturbance rejection	Eliminates residual steady-state tilt under constant bias while keeping the pole-placement structure.	Additional integral state increases tuning complexity and may reduce robustness if poorly placed.	Comparison case against LQI.
LQR	Optimal local stabilization	Systematic design through Bryson's rule; balances cart motion, angle regulation, and voltage effort.	Pure LQR cannot fully reject constant disturbances; steady-state error remains under bias torque.	Optimal stabilization baseline.
LQI	Optimal stabilization with integral action	Rejects constant disturbances and drives the steady-state angle error back to zero; suitable for biased hardware.	Requires careful weighting of the integral state to avoid excessive oscillation or control effort.	Best local balancing controller.
Kalman Filter / LQE	State estimation	Provides smoother estimates of unmeasured velocity states and avoids noisy numerical differentiation of encoder data.	Not yet cleanly validated in the loop; performance depends on the selected process and measurement noise covariance matrices.	Future estimator candidate.
CasADi + TV-LQR	Nonlinear lift-up maneuver	Handles the large nonlinear transition from the mechanical stop to the LQI capture region while respecting cart and voltage limits.	More complex and mainly validated in simulation; requires accurate nonlinear modeling and reliable switching logic.	Advanced nonlinear maneuver strategy.

The comparison follows the project progression: frequency-based control builds the practical cascade, state-space control handles the coupled dynamics more directly, and nonlinear trajectory control is needed for lift-up because local linear controllers apply only near the horizontal equilibrium.

10. Conclusion

This project developed a complete control workflow for the Quanser IP02 cart and Seesaw-E system: analytical modeling, parameter identification, controller synthesis, simulation validation, and hardware testing. The nonlinear equations were derived with Euler–Lagrange methods and linearized around the horizontal equilibrium; friction, deadzone, backlash, and sensor noise then required model refinement and compensation before deployment.

The experiments confirmed that the Seesaw is strongly coupled and non-minimum-phase, so the cart and body angle cannot be treated as independent SISO systems. Classical frequency-based control was useful for intuition and cart validation, while state-space control better handled the coupled balancing task. Integral action was essential for rejecting the steady-state error introduced by the 100 g asymmetric bias load.

The lift-up task showed the limitation of local linear controllers, since they are valid only near the horizontal equilibrium. For this reason, a hybrid nonlinear strategy was proposed: generate a feasible lift-up trajectory, track it with TV-LQR, and switch to the stationary balancing controller once the state enters the safe capture region.

Gear clicking, friction, degraded track behavior, encoder noise, sign anomalies, and QUARC/cache issues required repeated retuning and validation. These difficulties underline the value of conservative safety checks and structured MATLAB/Simulink scripts. Overall, the project met the project objectives: modeling, identification, controller design, implementation, validation, and critical comparison.

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- [2] S. Bruni and G. Bucca (2024). *Dynamics of Mechanical Systems* [Lecture notes]. Politecnico di Milano.